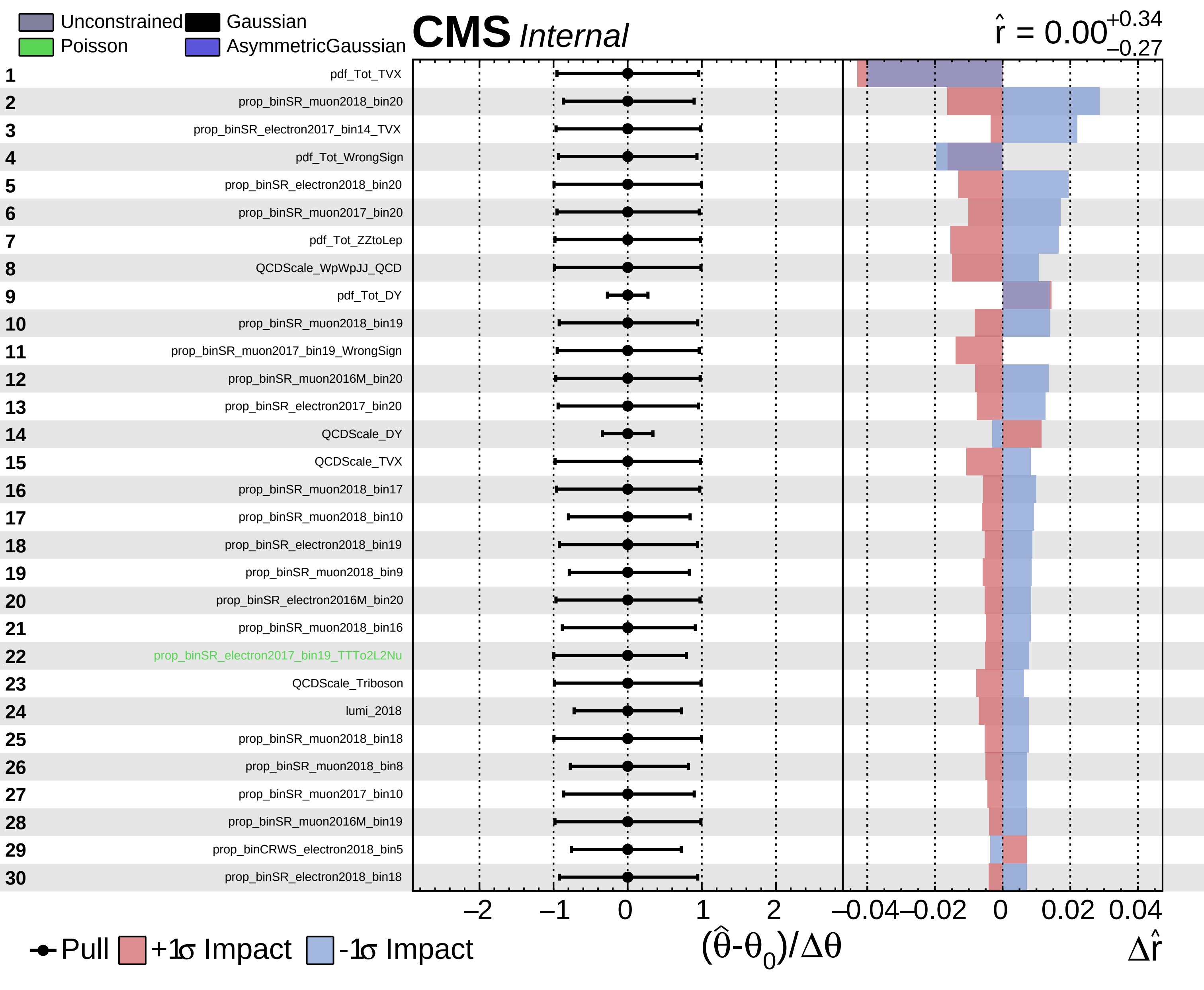
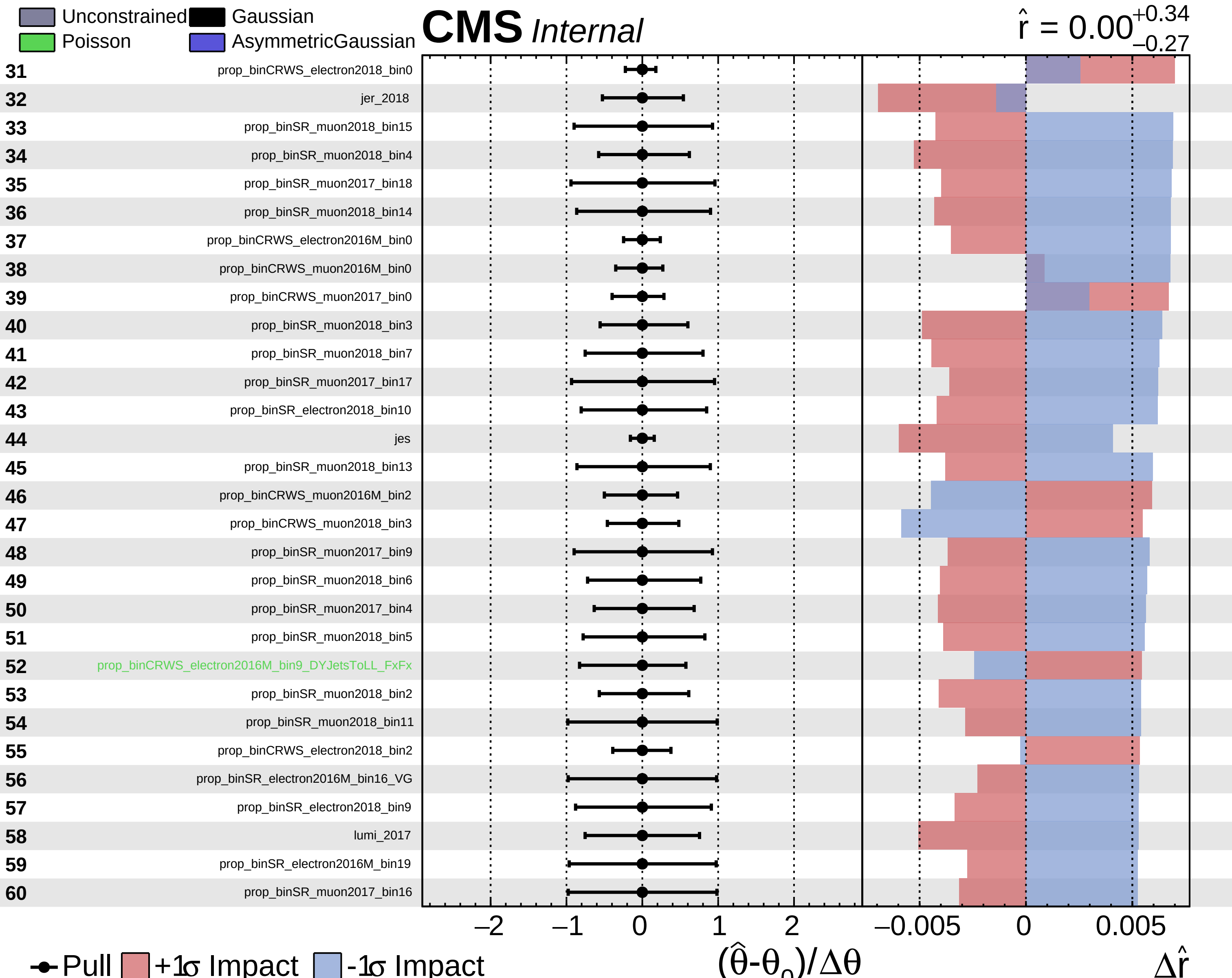
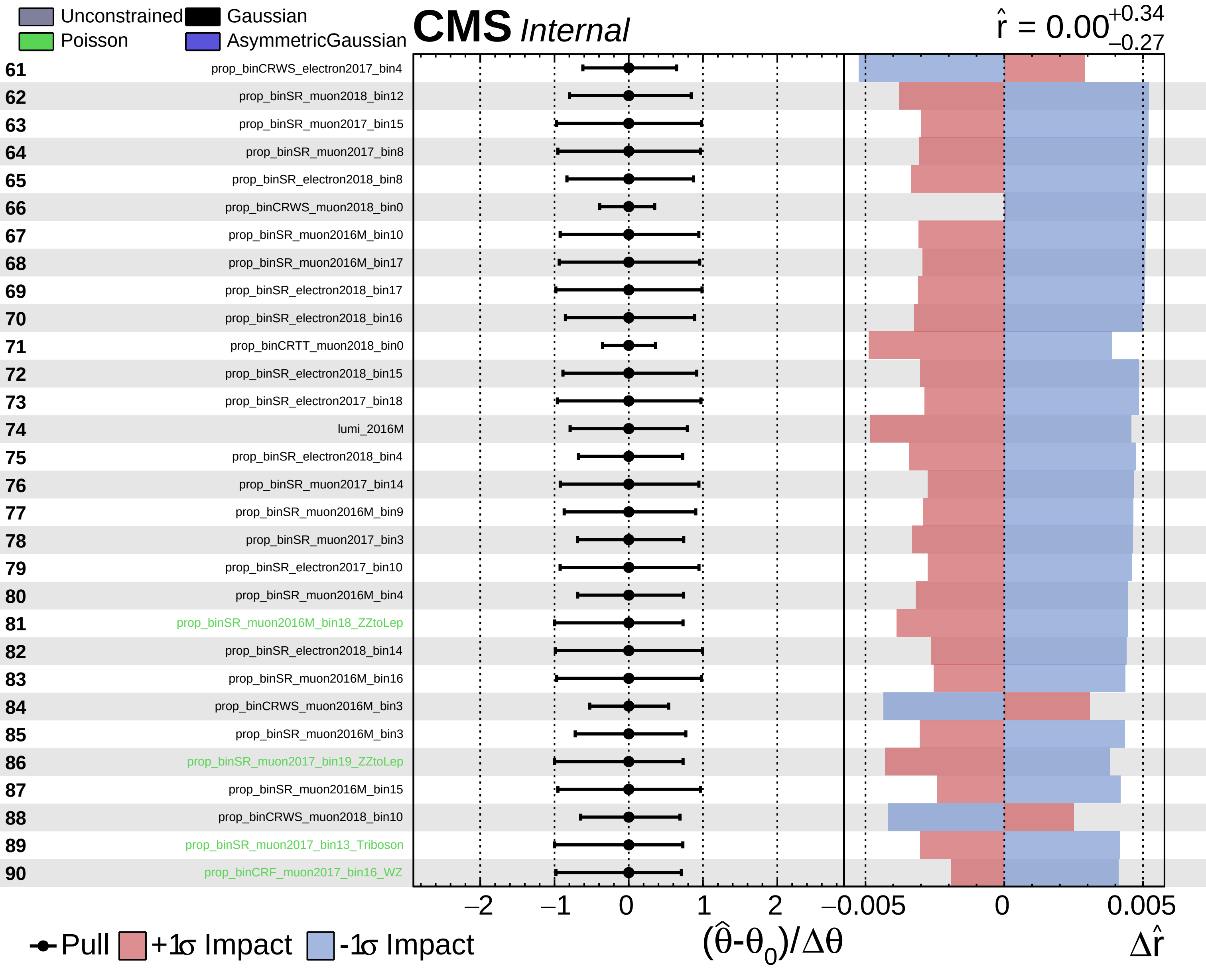


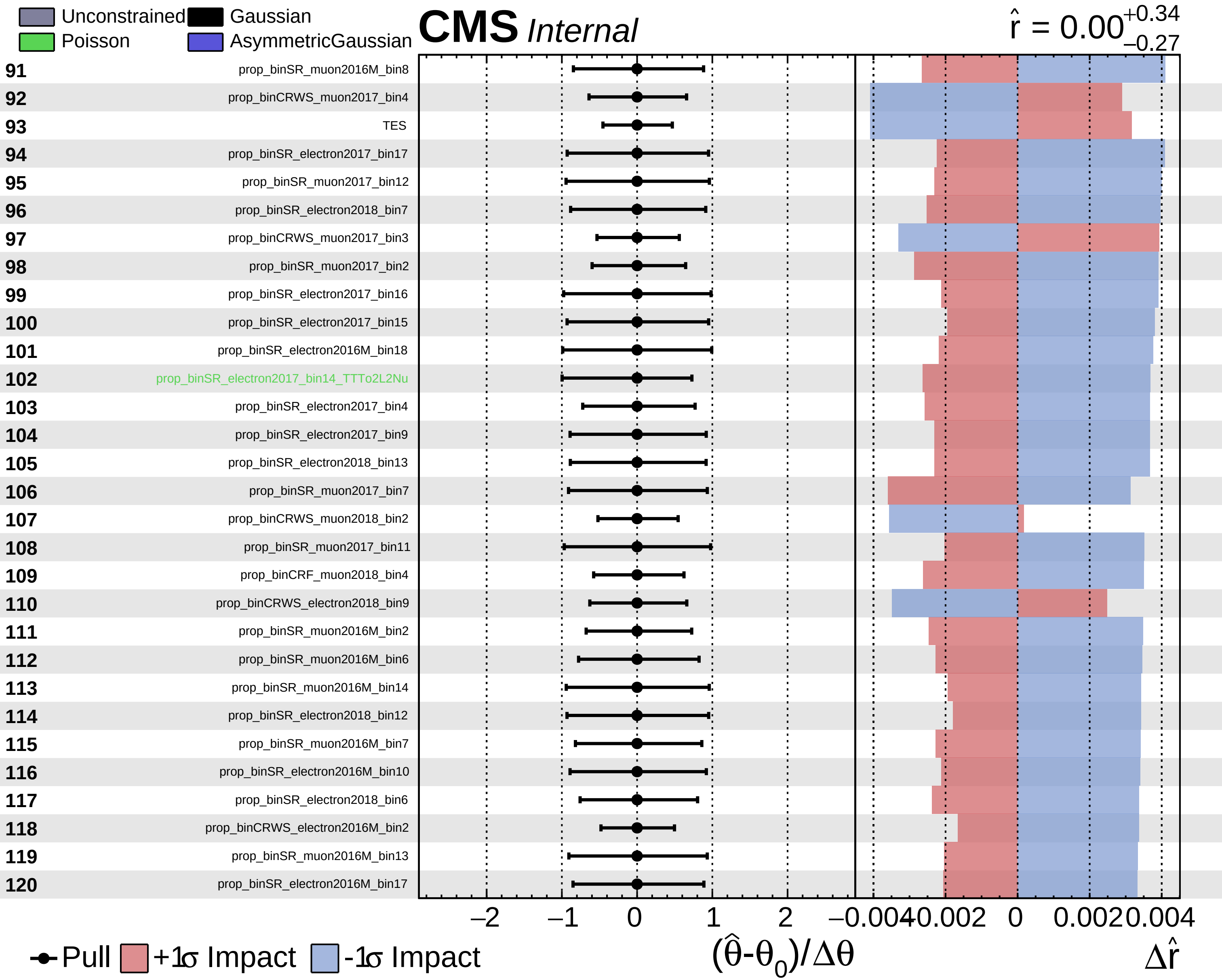
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
1 pdf_Tot_TVX
2 prop_binSR_muon2018_bin20
3 prop_binSR_electron2017_bin14_TVX
4 pdf_Tot_WrongSign
5 prop_binSR_electron2018_bin20
6 prop_binSR_muon2017_bin20
7 pdf_Tot_ZZtoLep
8 QCDScale_WpWpJJ_QCD
9 pdf_Tot_DY
10 prop_binSR_muon2018_bin19
11 prop_binSR_muon2017_bin19_WrongSign
12 prop_binSR_muon2016M_bin20
13 prop_binSR_electron2017_bin20
14 QCDScale_DY
15 QCDScale_TVX
16 prop_binSR_muon2018_bin17
17 prop_binSR_muon2018_bin10
18 prop_binSR_electron2018_bin19
19 prop_binSR_muon2018_bin9
20 prop_binSR_electron2016M_bin20
21 prop_binSR_muon2018_bin16
22 prop_binSR_electron2017_bin19_TTTo2L2Nu
23 QCDScale_Triboson
24 lumi_2018
25 prop_binSR_muon2018_bin18
26 prop_binSR_muon2018_bin8
27 prop_binSR_muon2017_bin10
28 prop_binSR_muon2016M_bin19
29 prop_binCRWS_electron2018_bin5
30 prop_binSR_electron2018_bin18
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



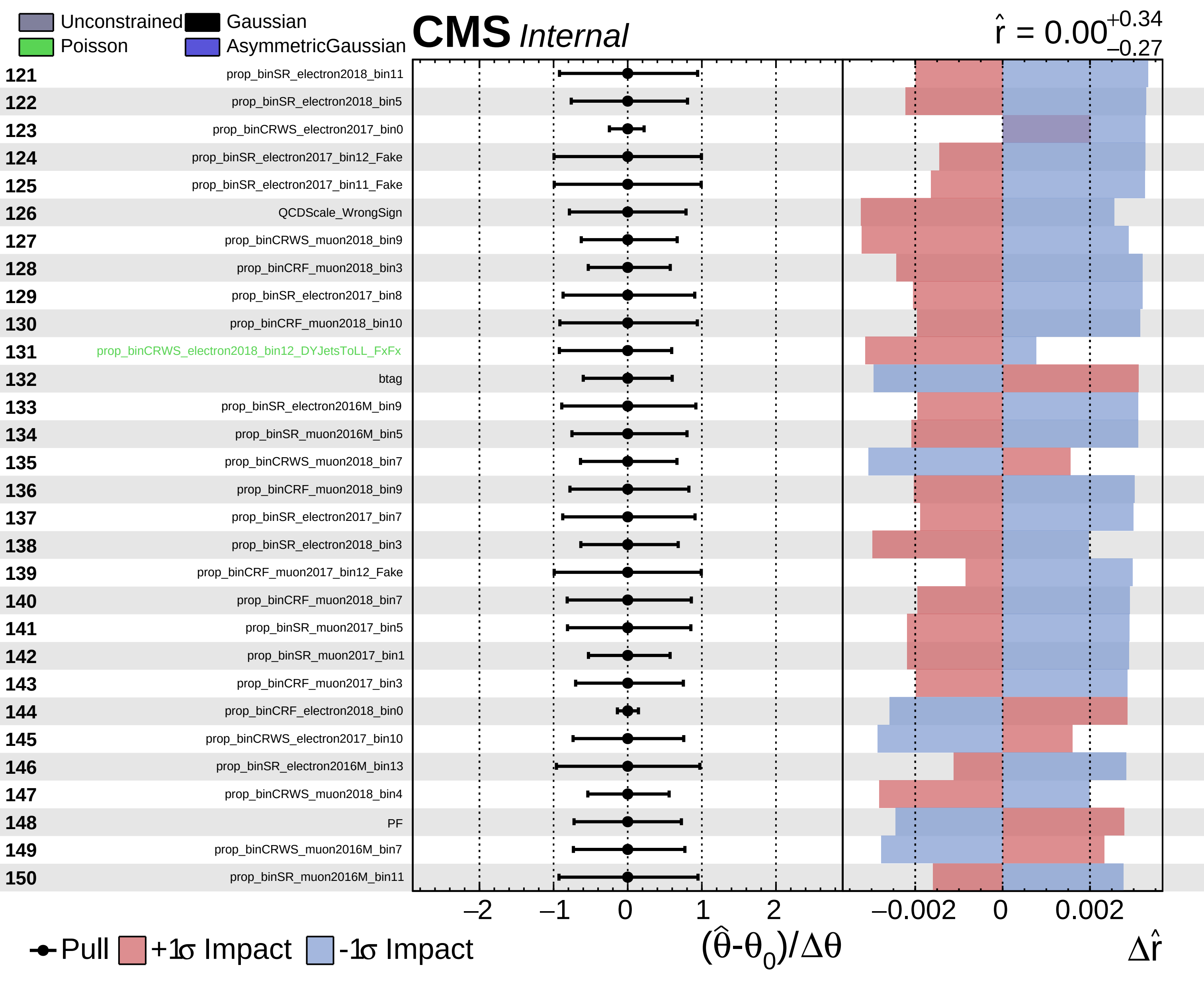
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
31 prop_binCRWS_electron2018_bin0
32 jer_2018
33 prop_binSR_muon2018_bin15
34 prop_binSR_muon2018_bin4
35 prop_binSR_muon2017_bin18
36 prop_binSR_muon2018_bin14
37 prop_binCRWS_electron2016M_bin0
38 prop_binCRWS_muon2016M_bin0
39 prop_binCRWS_muon2017_bin0
40 prop_binSR_muon2018_bin3
41 prop_binSR_muon2018_bin7
42 prop_binSR_muon2017_bin17
43 prop_binSR_electron2018_bin10
44 jes
45 prop_binSR_muon2018_bin13
46 prop_binCRWS_muon2016M_bin2
47 prop_binCRWS_muon2018_bin3
48 prop_binSR_muon2017_bin9
49 prop_binSR_muon2018_bin6
50 prop_binSR_muon2017_bin4
51 prop_binSR_muon2018_bin5
52 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
53 prop_binSR_muon2018_bin2
54 prop_binSR_muon2018_bin11
55 prop_binCRWS_electron2018_bin2
56 prop_binSR_electron2016M_bin16_VG
57 prop_binSR_electron2018_bin9
58 lumi_2017
59 prop_binSR_electron2016M_bin19
60 prop_binSR_muon2017_bin16
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

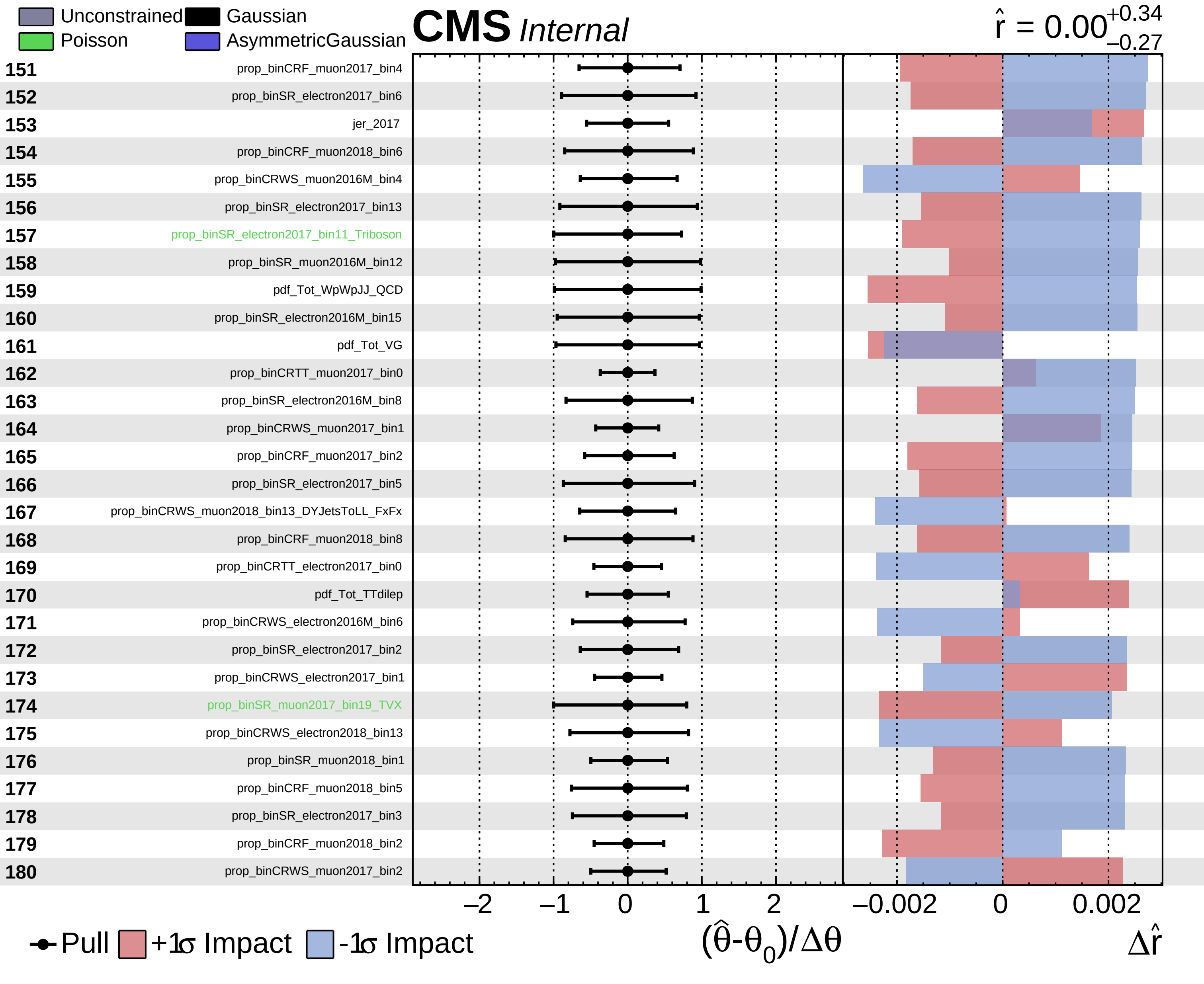


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
61 prop_binCRWS_electron2017_bin4
62 prop_binSR_muon2018_bin12
63 prop_binSR_muon2017_bin15
64 prop_binSR_muon2017_bin8
65 prop_binSR_electron2018_bin8
66 prop_binCRWS_muon2018_bin0
67 prop_binSR_muon2016M_bin10
68 prop_binSR_muon2016M_bin17
69 prop_binSR_electron2018_bin17
70 prop_binSR_electron2018_bin16
71 prop_binCRTT_muon2018_bin0
72 prop_binSR_electron2018_bin15
73 prop_binSR_electron2017_bin18
74 lumi_2016M
75 prop_binSR_electron2018_bin4
76 prop_binSR_muon2017_bin14
77 prop_binSR_muon2016M_bin9
78 prop_binSR_muon2017_bin3
79 prop_binSR_electron2017_bin10
80 prop_binSR_muon2016M_bin4
81 prop_binSR_muon2016M_bin18_ZZtoLep
82 prop_binSR_electron2018_bin14
83 prop_binSR_muon2016M_bin16
84 prop_binCRWS_muon2016M_bin3
85 prop_binSR_muon2016M_bin3
86 prop_binSR_muon2017_bin19_ZZtoLep
87 prop_binSR_muon2016M_bin15
88 prop_binCRWS_muon2018_bin10
89 prop_binSR_muon2017_bin13_Triboson
90 prop_binCRF_muon2017_bin16_WZ
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

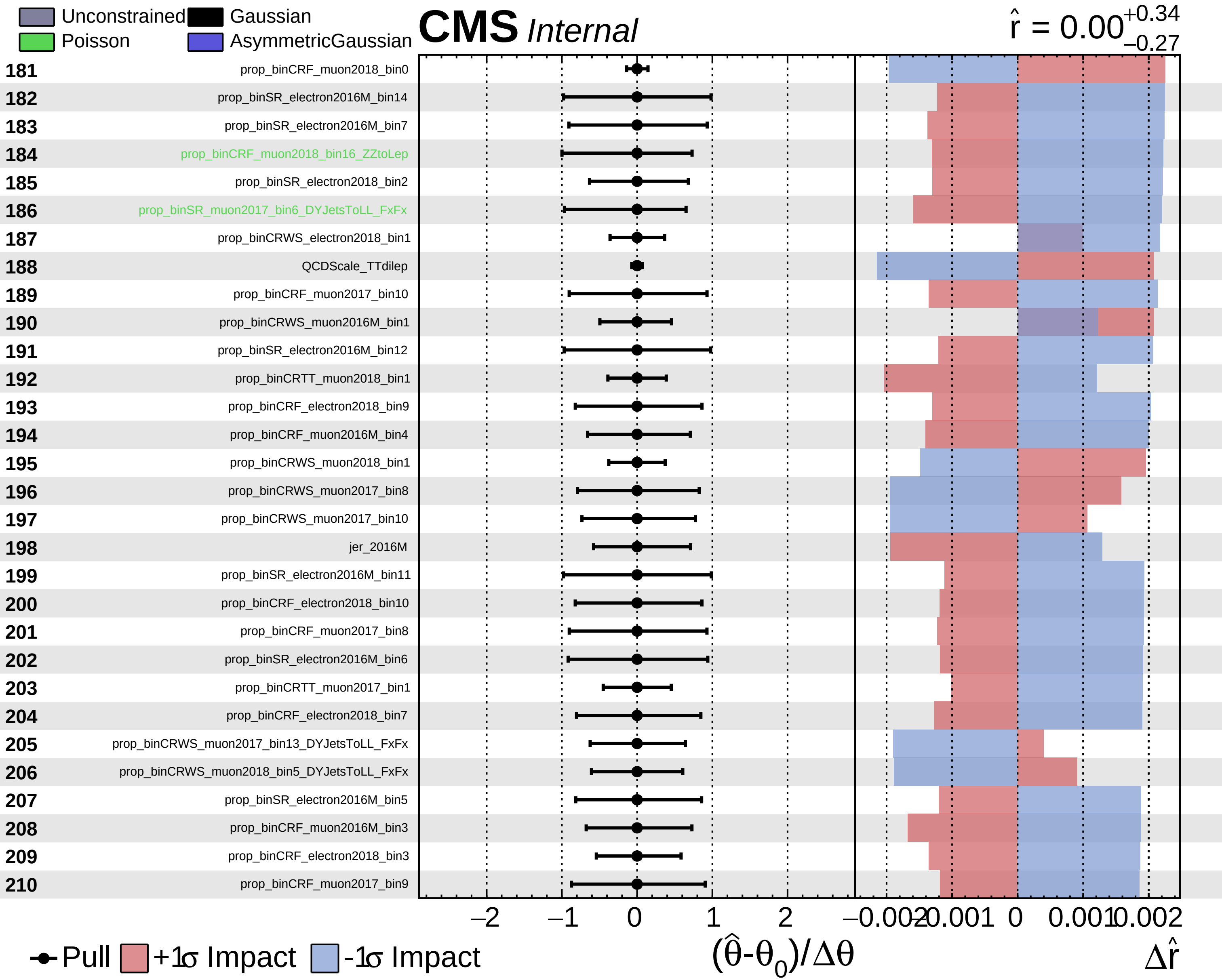


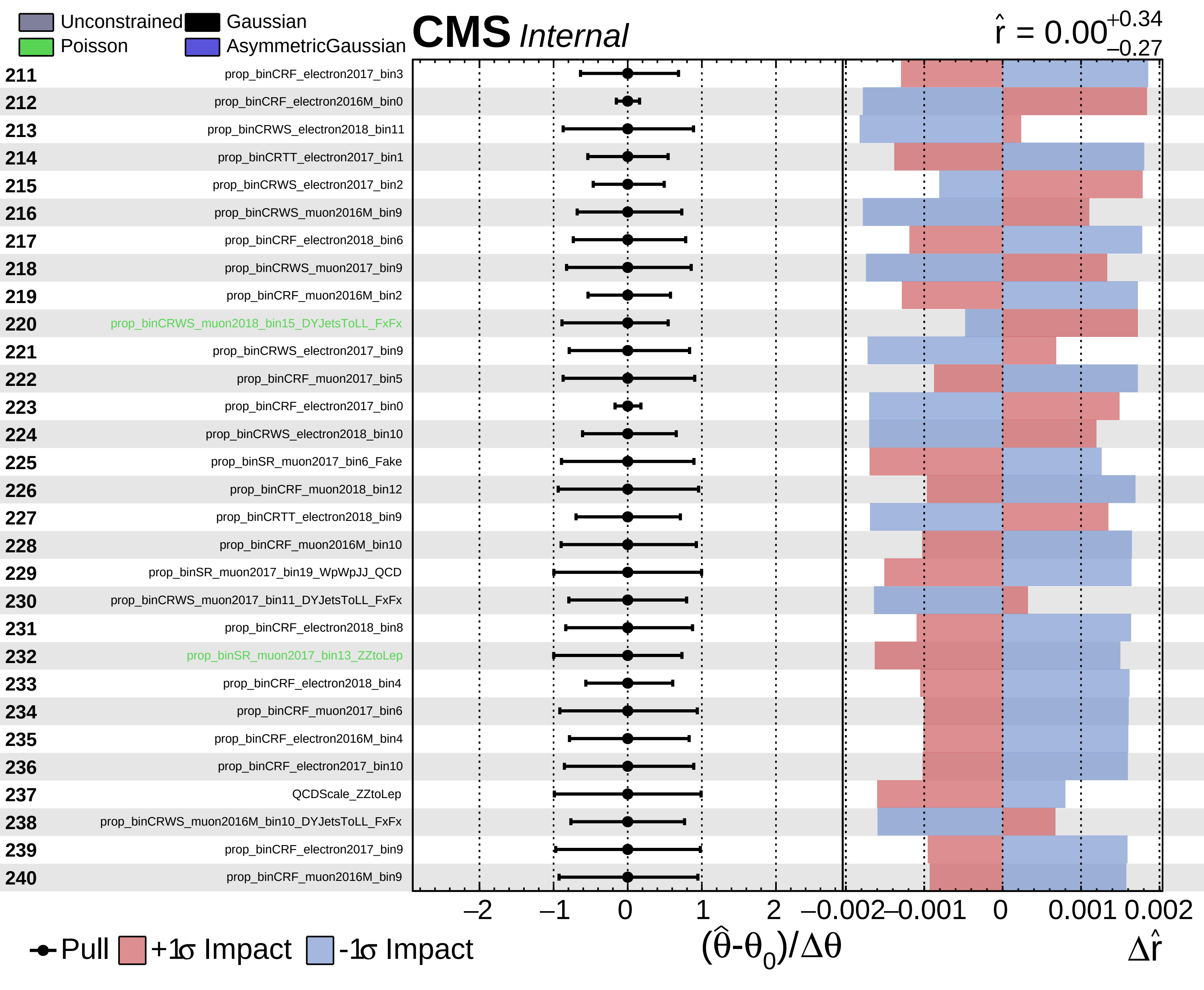
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
91 prop_binSR_muon2016M_bin8
92 prop_binCRWS_muon2017_bin4
93 TES
94 prop_binSR_electron2017_bin17
95 prop_binSR_muon2017_bin12
96 prop_binSR_electron2018_bin7
97 prop_binCRWS_muon2017_bin3
98 prop_binSR_muon2017_bin2
99 prop_binSR_electron2017_bin16
100 prop_binSR_electron2017_bin15
101 prop_binSR_electron2016M_bin18
102 prop_binSR_electron2017_bin14_TTTo2L2Nu
103 prop_binSR_electron2017_bin4
104 prop_binSR_electron2017_bin9
105 prop_binSR_electron2018_bin13
106 prop_binSR_muon2017_bin7
107 prop_binCRWS_muon2018_bin2
108 prop_binSR_muon2017_bin11
109 prop_binCRF_muon2018_bin4
110 prop_binCRWS_electron2018_bin9
111 prop_binSR_muon2016M_bin2
112 prop_binSR_muon2016M_bin6
113 prop_binSR_muon2016M_bin14
114 prop_binSR_electron2018_bin12
115 prop_binSR_muon2016M_bin7
116 prop_binSR_electron2016M_bin10
117 prop_binSR_electron2018_bin6
118 prop_binCRWS_electron2016M_bin2
119 prop_binSR_muon2016M_bin13
120 prop_binSR_electron2016M_bin17
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
121 prop_binSR_electron2018_bin11
122 prop_binSR_electron2018_bin5
123 prop_binCRWS_electron2017_bin0
124 prop_binSR_electron2017_bin12_Fake
125 prop_binSR_electron2017_bin11_Fake
126 QCDScale_WrongSign
127 prop_binCRWS_muon2018_bin9
128 prop_binCRF_muon2018_bin3
129 prop_binSR_electron2017_bin8
130 prop_binCRF_muon2018_bin10
131 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
132 btag
133 prop_binSR_electron2016M_bin9
134 prop_binSR_muon2016M_bin5
135 prop_binCRWS_muon2018_bin7
136 prop_binCRF_muon2018_bin9
137 prop_binSR_electron2017_bin7
138 prop_binSR_electron2018_bin3
139 prop_binCRF_muon2017_bin12_Fake
140 prop_binCRF_muon2018_bin7
141 prop_binSR_muon2017_bin5
142 prop_binSR_muon2017_bin1
143 prop_binCRF_muon2017_bin3
144 prop_binCRF_electron2018_bin0
145 prop_binCRWS_electron2017_bin10
146 prop_binSR_electron2016M_bin13
147 prop_binCRWS_muon2018_bin4
148 PF
149 prop_binCRWS_muon2016M_bin7
150 prop_binSR_muon2016M_bin11
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

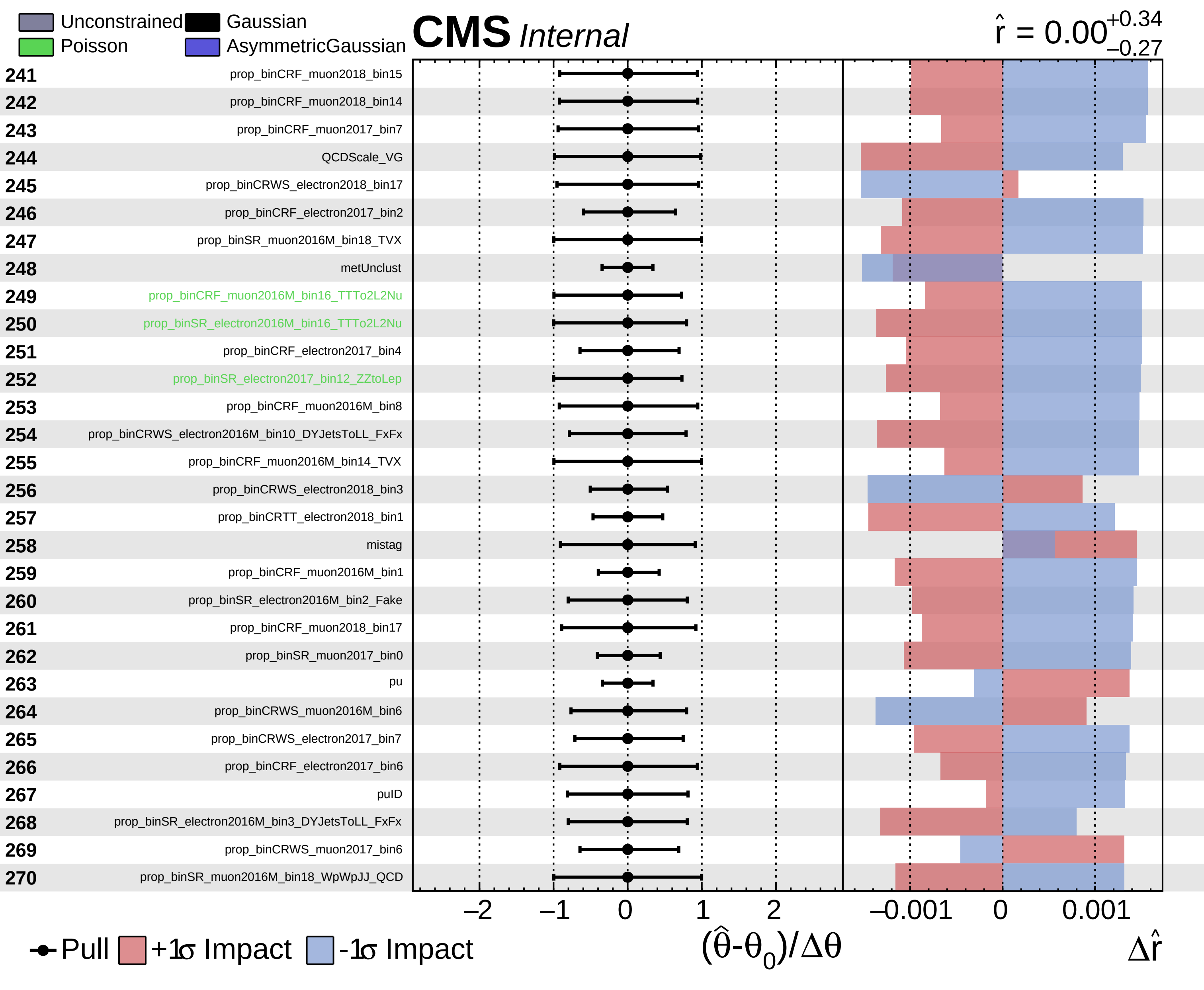


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
151 prop_binCRF_muon2017_bin4
152 prop_binSR_electron2017_bin6
153 jer_2017
154 prop_binCRF_muon2018_bin6
155 prop_binCRWS_muon2016M_bin4
156 prop_binSR_electron2017_bin13
157 prop_binSR_electron2017_bin11_Triboson
158 prop_binSR_muon2016M_bin12
159 pdf_Tot_WpWpJJ_QCD
160 prop_binSR_electron2016M_bin15
161 pdf_Tot_VG
162 prop_binCRTT_muon2017_bin0
163 prop_binSR_electron2016M_bin8
164 prop_binCRWS_muon2017_bin1
165 prop_binCRF_muon2017_bin2
166 prop_binSR_electron2017_bin5
167 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
168 prop_binCRF_muon2018_bin8
169 prop_binCRTT_electron2017_bin0
170 pdf_Tot_TTdilep
171 prop_binCRWS_electron2016M_bin6
172 prop_binSR_electron2017_bin2
173 prop_binCRWS_electron2017_bin1
174 prop_binSR_muon2017_bin19_TVX
175 prop_binCRWS_electron2018_bin13
176 prop_binSR_muon2018_bin1
177 prop_binCRF_muon2018_bin5
178 prop_binSR_electron2017_bin3
179 prop_binCRF_muon2018_bin2
180 prop_binCRWS_muon2017_bin2
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

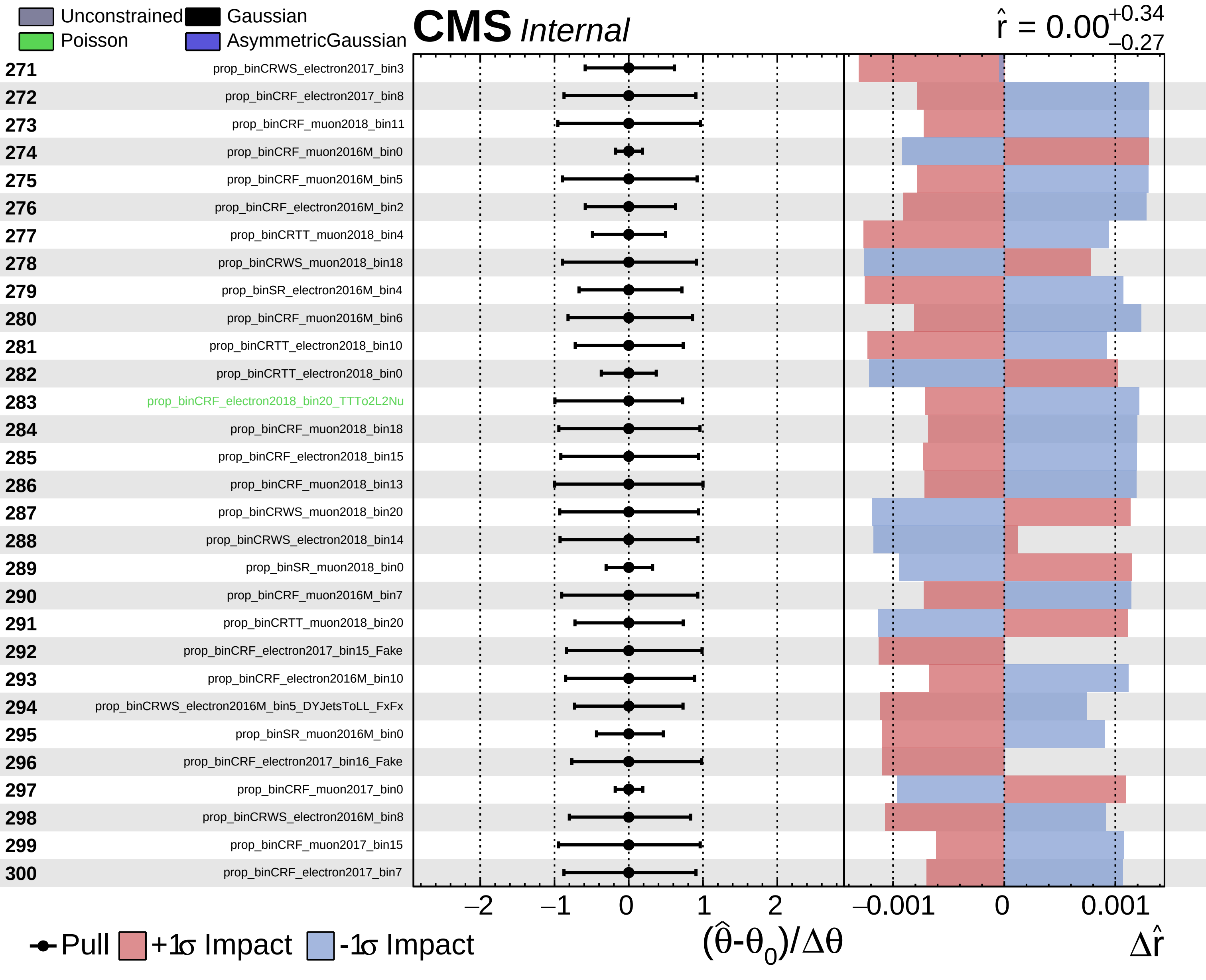

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
181 prop_binCRF_muon2018_bin0
182 prop_binSR_electron2016M_bin14
183 prop_binSR_electron2016M_bin7
184 prop_binCRF_muon2018_bin16_ZZtoLep
185 prop_binSR_electron2018_bin2
186 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
187 prop_binCRWS_electron2018_bin1
188 QCDScale_TTdilep
189 prop_binCRF_muon2017_bin10
190 prop_binCRWS_muon2016M_bin1
191 prop_binSR_electron2016M_bin12
192 prop_binCRTT_muon2018_bin1
193 prop_binCRF_electron2018_bin9
194 prop_binCRF_muon2016M_bin4
195 prop_binCRWS_muon2018_bin1
196 prop_binCRWS_muon2017_bin8
197 prop_binCRWS_muon2017_bin10
198 jer_2016M
199 prop_binSR_electron2016M_bin11
200 prop_binCRF_electron2018_bin10
201 prop_binCRF_muon2017_bin8
202 prop_binSR_electron2016M_bin6
203 prop_binCRTT_muon2017_bin1
204 prop_binCRF_electron2018_bin7
205 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
206 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
207 prop_binSR_electron2016M_bin5
208 prop_binCRF_muon2016M_bin3
209 prop_binCRF_electron2018_bin3
210 prop_binCRF_muon2017_bin9
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



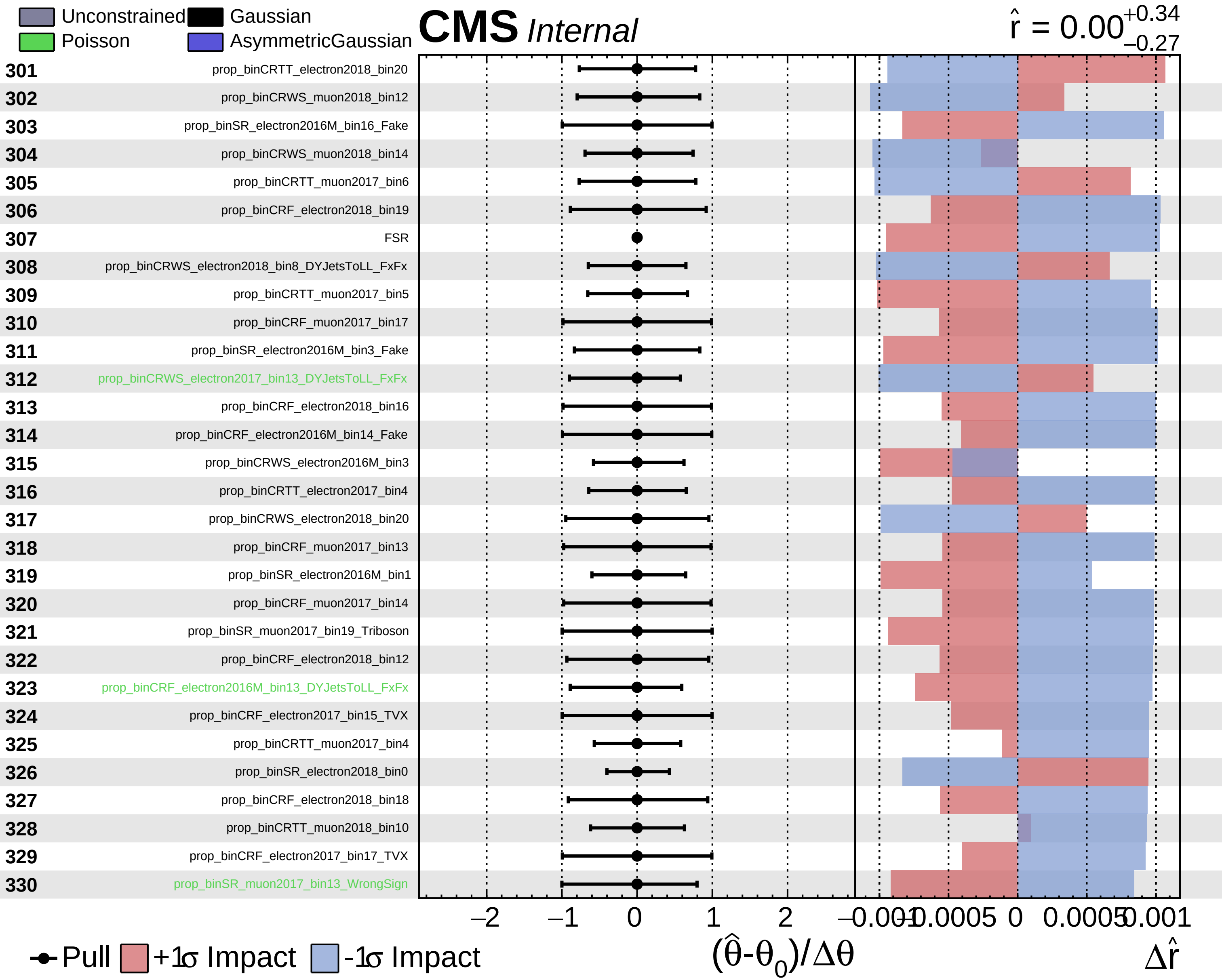
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
211 prop_binCRF_electron2017_bin3
212 prop_binCRF_electron2016M_bin0
213 prop_binCRWS_electron2018_bin11
214 prop_binCRTT_electron2017_bin1
215 prop_binCRWS_electron2017_bin2
216 prop_binCRWS_muon2016M_bin9
217 prop_binCRF_electron2018_bin6
218 prop_binCRWS_muon2017_bin9
219 prop_binCRF_muon2016M_bin2
220 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
221 prop_binCRWS_electron2017_bin9
222 prop_binCRF_muon2017_bin5
223 prop_binCRF_electron2017_bin0
224 prop_binCRWS_electron2018_bin10
225 prop_binSR_muon2017_bin6_Fake
226 prop_binCRF_muon2018_bin12
227 prop_binCRTT_electron2018_bin9
228 prop_binCRF_muon2016M_bin10
229 prop_binSR_muon2017_bin19_WpWpJJ_QCD
230 prop_binCRWS_muon2017_bin11_DYJetsToLL_FxFx
231 prop_binCRF_electron2018_bin8
232 prop_binSR_muon2017_bin13_ZZtoLep
233 prop_binCRF_electron2018_bin4
234 prop_binCRF_muon2017_bin6
235 prop_binCRF_electron2016M_bin4
236 prop_binCRF_electron2017_bin10
237 QCDScale_ZZtoLep
238 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
239 prop_binCRF_electron2017_bin9
240 prop_binCRF_muon2016M_bin9
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



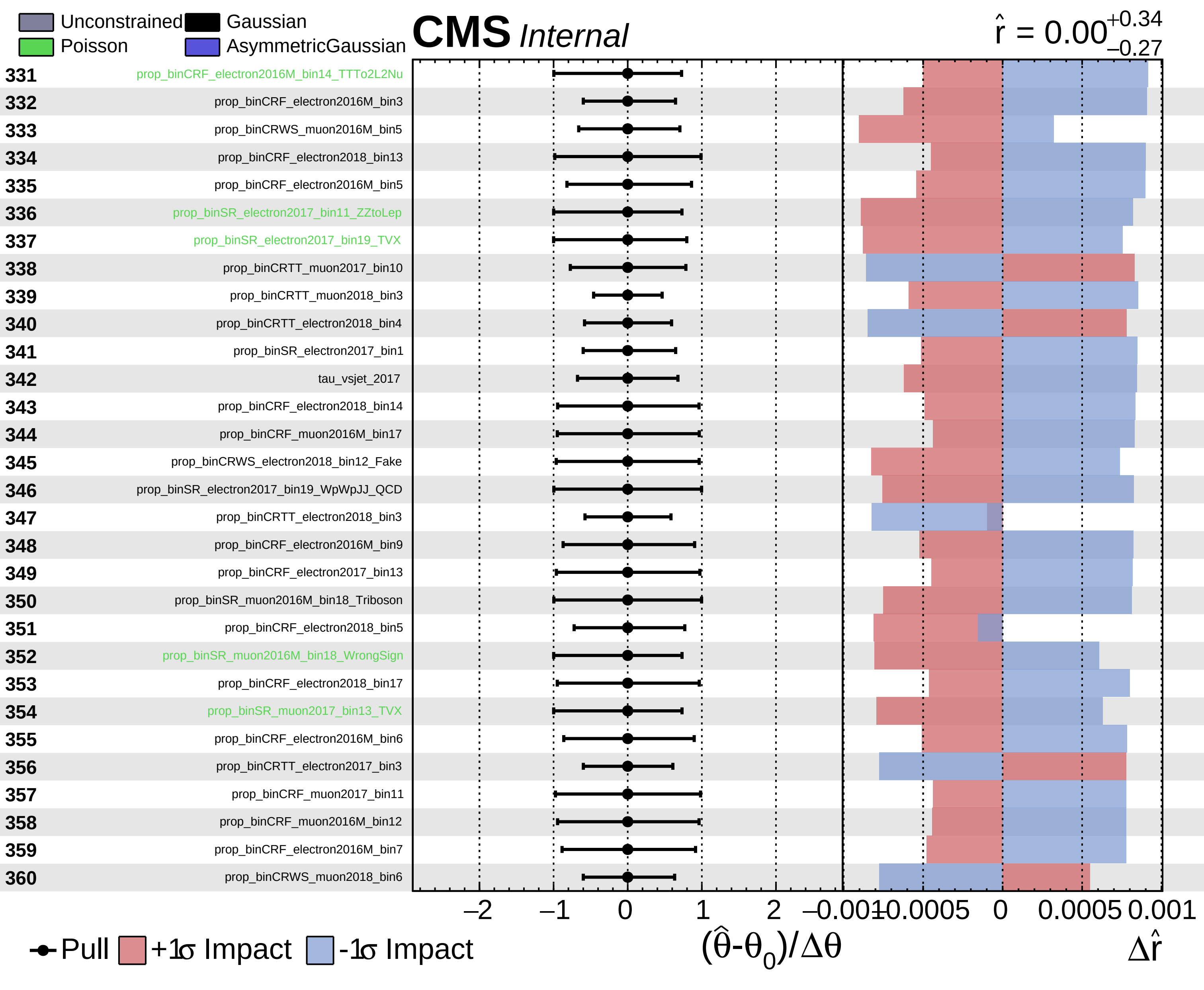
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
241 prop_binCRF_muon2018_bin15
242 prop_binCRF_muon2018_bin14
243 prop_binCRF_muon2017_bin7
244 QCDScale_VG
245 prop_binCRWS_electron2018_bin17
246 prop_binCRF_electron2017_bin2
247 prop_binSR_muon2016M_bin18_TVX
248 metUnclust
249 prop_binCRF_muon2016M_bin16_TTTo2L2Nu
250 prop_binSR_electron2016M_bin16_TTTo2L2Nu
251 prop_binCRF_electron2017_bin4
252 prop_binSR_electron2017_bin12_ZZtoLep
253 prop_binCRF_muon2016M_bin8
254 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
255 prop_binCRF_muon2016M_bin14_TVX
256 prop_binCRWS_electron2018_bin3
257 prop_binCRTT_electron2018_bin1
258 mistag
259 prop_binCRF_muon2016M_bin1
260 prop_binSR_electron2016M_bin2_Fake
261 prop_binCRF_muon2018_bin17
262 prop_binSR_muon2017_bin0
263 pu
264 prop_binCRWS_muon2016M_bin6
265 prop_binCRWS_electron2017_bin7
266 prop_binCRF_electron2017_bin6
267 puID
268 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
269 prop_binCRWS_muon2017_bin6
270 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

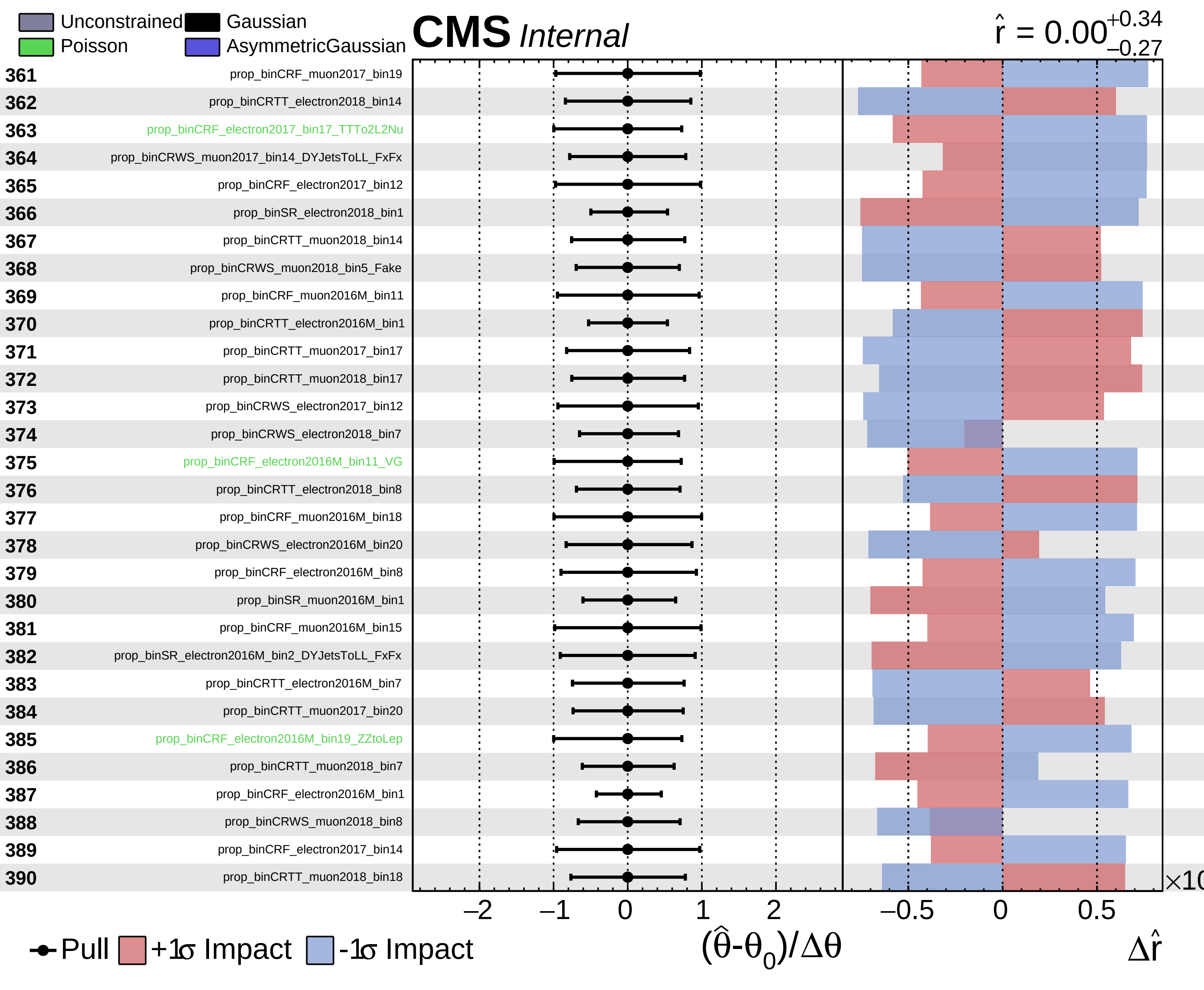


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
271 prop_binCRWS_electron2017_bin3
272 prop_binCRF_electron2017_bin8
273 prop_binCRF_muon2018_bin11
274 prop_binCRF_muon2016M_bin0
275 prop_binCRF_muon2016M_bin5
276 prop_binCRF_electron2016M_bin2
277 prop_binCRTT_muon2018_bin4
278 prop_binCRWS_muon2018_bin18
279 prop_binSR_electron2016M_bin4
280 prop_binCRF_muon2016M_bin6
281 prop_binCRTT_electron2018_bin10
282 prop_binCRTT_electron2018_bin0
283 prop_binCRF_electron2018_bin20_TTTo2L2Nu
284 prop_binCRF_muon2018_bin18
285 prop_binCRF_electron2018_bin15
286 prop_binCRF_muon2018_bin13
287 prop_binCRWS_muon2018_bin20
288 prop_binCRWS_electron2018_bin14
289 prop_binSR_muon2018_bin0
290 prop_binCRF_muon2016M_bin7
291 prop_binCRTT_muon2018_bin20
292 prop_binCRF_electron2017_bin15_Fake
293 prop_binCRF_electron2016M_bin10
294 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
295 prop_binSR_muon2016M_bin0
296 prop_binCRF_electron2017_bin16_Fake
297 prop_binCRF_muon2017_bin0
298 prop_binCRWS_electron2016M_bin8
299 prop_binCRF_muon2017_bin15
300 prop_binCRF_electron2017_bin7
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

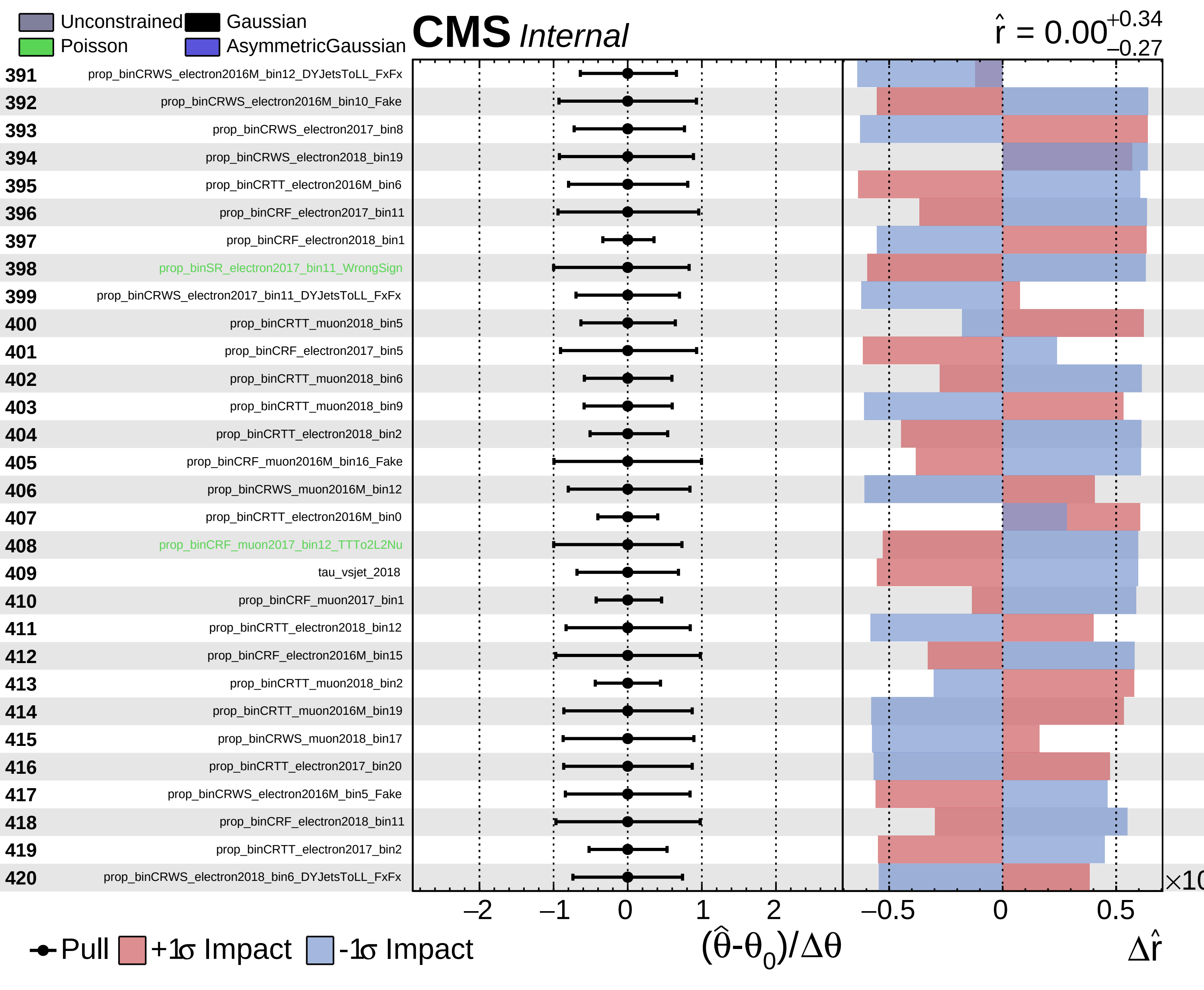


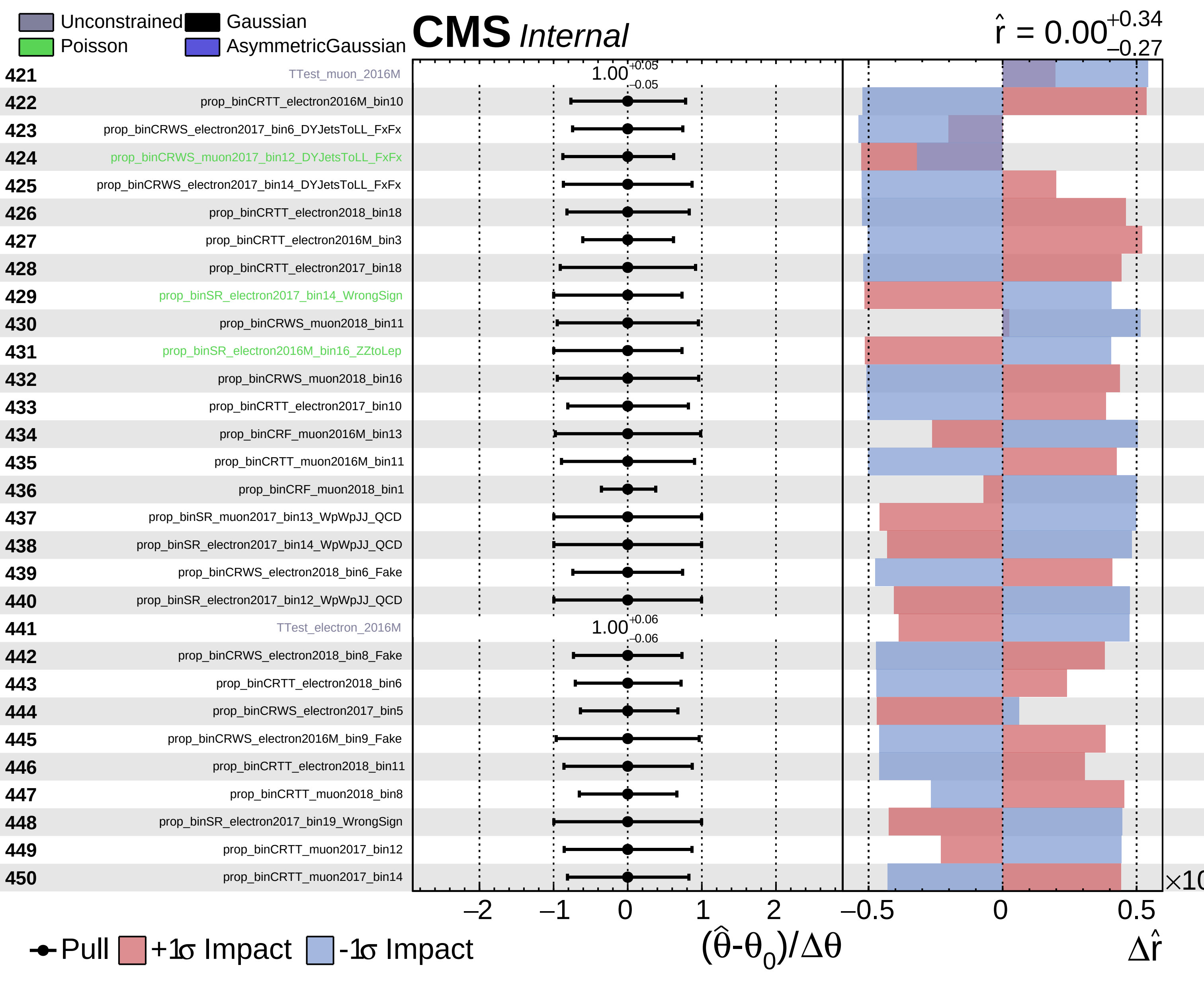
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
301 prop_binCRTT_electron2018_bin20
302 prop_binCRWS_muon2018_bin12
303 prop_binSR_electron2016M_bin16_Fake
304 prop_binCRWS_muon2018_bin14
305 prop_binCRTT_muon2017_bin6
306 prop_binCRF_electron2018_bin19
307 FSR
308 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
309 prop_binCRTT_muon2017_bin5
310 prop_binCRF_muon2017_bin17
311 prop_binSR_electron2016M_bin3_Fake
312 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
313 prop_binCRF_electron2018_bin16
314 prop_binCRF_electron2016M_bin14_Fake
315 prop_binCRWS_electron2016M_bin3
316 prop_binCRTT_electron2017_bin4
317 prop_binCRWS_electron2018_bin20
318 prop_binCRF_muon2017_bin13
319 prop_binSR_electron2016M_bin1
320 prop_binCRF_muon2017_bin14
321 prop_binSR_muon2017_bin19_Triboson
322 prop_binCRF_electron2018_bin12
323 prop_binCRF_electron2016M_bin13_DYJetsToLL_FxFx
324 prop_binCRF_electron2017_bin15_TVX
325 prop_binCRTT_muon2017_bin4
326 prop_binSR_electron2018_bin0
327 prop_binCRF_electron2018_bin18
328 prop_binCRTT_muon2018_bin10
329 prop_binCRF_electron2017_bin17_TVX
330 prop_binSR_muon2017_bin13_WrongSign
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
331 prop_binCRF_electron2016M_bin14_TTTo2L2Nu
332 prop_binCRF_electron2016M_bin3
333 prop_binCRWS_muon2016M_bin5
334 prop_binCRF_electron2018_bin13
335 prop_binCRF_electron2016M_bin5
336 prop_binSR_electron2017_bin11_ZZtoLep
337 prop_binSR_electron2017_bin19_TVX
338 prop_binCRTT_muon2017_bin10
339 prop_binCRTT_muon2018_bin3
340 prop_binCRTT_electron2018_bin4
341 prop_binSR_electron2017_bin1
342 tau_vsjet_2017
343 prop_binCRF_electron2018_bin14
344 prop_binCRF_muon2016M_bin17
345 prop_binCRWS_electron2018_bin12_Fake
346 prop_binSR_electron2017_bin19_WpWpJJ_QCD
347 prop_binCRTT_electron2018_bin3
348 prop_binCRF_electron2016M_bin9
349 prop_binCRF_electron2017_bin13
350 prop_binSR_muon2016M_bin18_Triboson
351 prop_binCRF_electron2018_bin5
352 prop_binSR_muon2016M_bin18_WrongSign
353 prop_binCRF_electron2018_bin17
354 prop_binSR_muon2017_bin13_TVX
355 prop_binCRF_electron2016M_bin6
356 prop_binCRTT_electron2017_bin3
357 prop_binCRF_muon2017_bin11
358 prop_binCRF_muon2016M_bin12
359 prop_binCRF_electron2016M_bin7
360 prop_binCRWS_muon2018_bin6
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

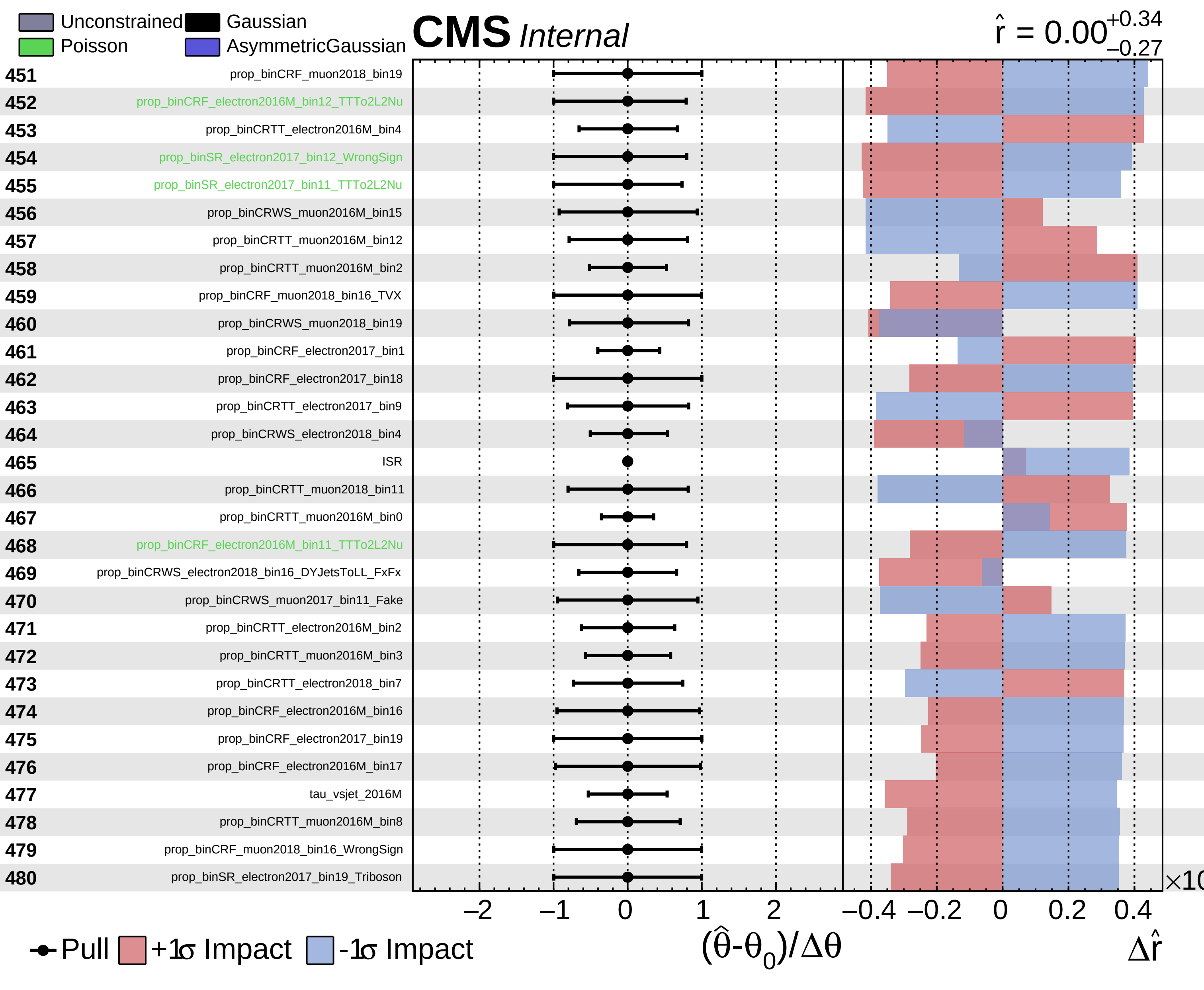


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
361 prop_binCRF_muon2017_bin19
362 prop_binCRTT_electron2018_bin14
363 prop_binCRF_electron2017_bin17_TTTo2L2Nu
364 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
365 prop_binCRF_electron2017_bin12
366 prop_binSR_electron2018_bin1
367 prop_binCRTT_muon2018_bin14
368 prop_binCRWS_muon2018_bin5_Fake
369 prop_binCRF_muon2016M_bin11
370 prop_binCRTT_electron2016M_bin1
371 prop_binCRTT_muon2017_bin17
372 prop_binCRTT_muon2018_bin17
373 prop_binCRWS_electron2017_bin12
374 prop_binCRWS_electron2018_bin7
375 prop_binCRF_electron2016M_bin11_VG
376 prop_binCRTT_electron2018_bin8
377 prop_binCRF_muon2016M_bin18
378 prop_binCRWS_electron2016M_bin20
379 prop_binCRF_electron2016M_bin8
380 prop_binSR_muon2016M_bin1
381 prop_binCRF_muon2016M_bin15
382 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
383 prop_binCRTT_electron2016M_bin7
384 prop_binCRTT_muon2017_bin20
385 prop_binCRF_electron2016M_bin19_ZZtoLep
386 prop_binCRTT_muon2018_bin7
387 prop_binCRF_electron2016M_bin1
388 prop_binCRWS_muon2018_bin8
389 prop_binCRF_electron2017_bin14
390 prop_binCRTT_muon2018_bin18
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

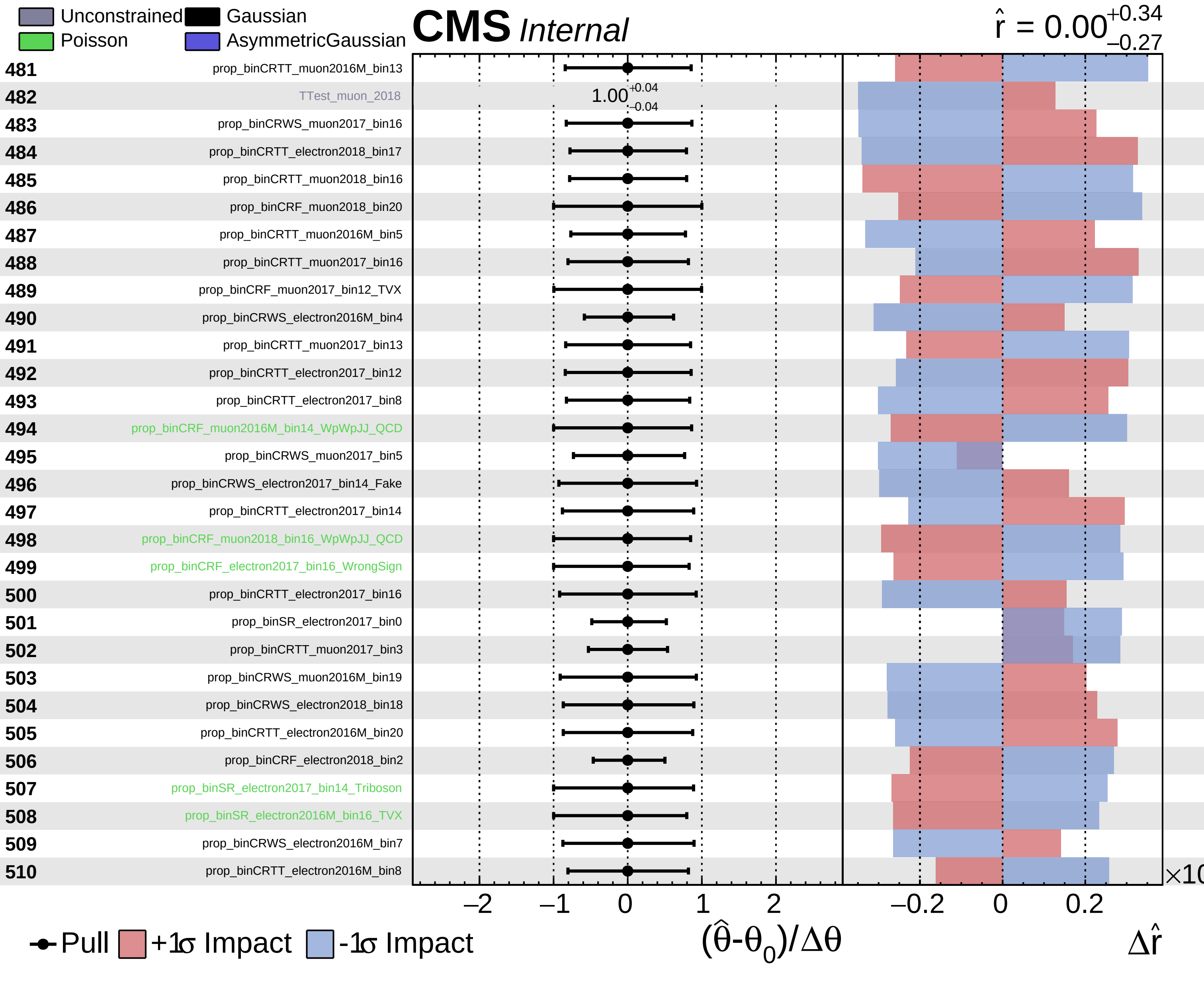

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
391 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
392 prop_binCRWS_electron2016M_bin10_Fake
393 prop_binCRWS_electron2017_bin8
394 prop_binCRWS_electron2018_bin19
395 prop_binCRTT_electron2016M_bin6
396 prop_binCRF_electron2017_bin11
397 prop_binCRF_electron2018_bin1
398 prop_binSR_electron2017_bin11_WrongSign
399 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
400 prop_binCRTT_muon2018_bin5
401 prop_binCRF_electron2017_bin5
402 prop_binCRTT_muon2018_bin6
403 prop_binCRTT_muon2018_bin9
404 prop_binCRTT_electron2018_bin2
405 prop_binCRF_muon2016M_bin16_Fake
406 prop_binCRWS_muon2016M_bin12
407 prop_binCRTT_electron2016M_bin0
408 prop_binCRF_muon2017_bin12_TTTo2L2Nu
409 tau_vsjet_2018
410 prop_binCRF_muon2017_bin1
411 prop_binCRTT_electron2018_bin12
412 prop_binCRF_electron2016M_bin15
413 prop_binCRTT_muon2018_bin2
414 prop_binCRTT_muon2016M_bin19
415 prop_binCRWS_muon2018_bin17
416 prop_binCRTT_electron2017_bin20
417 prop_binCRWS_electron2016M_bin5_Fake
418 prop_binCRF_electron2018_bin11
419 prop_binCRTT_electron2017_bin2
420 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

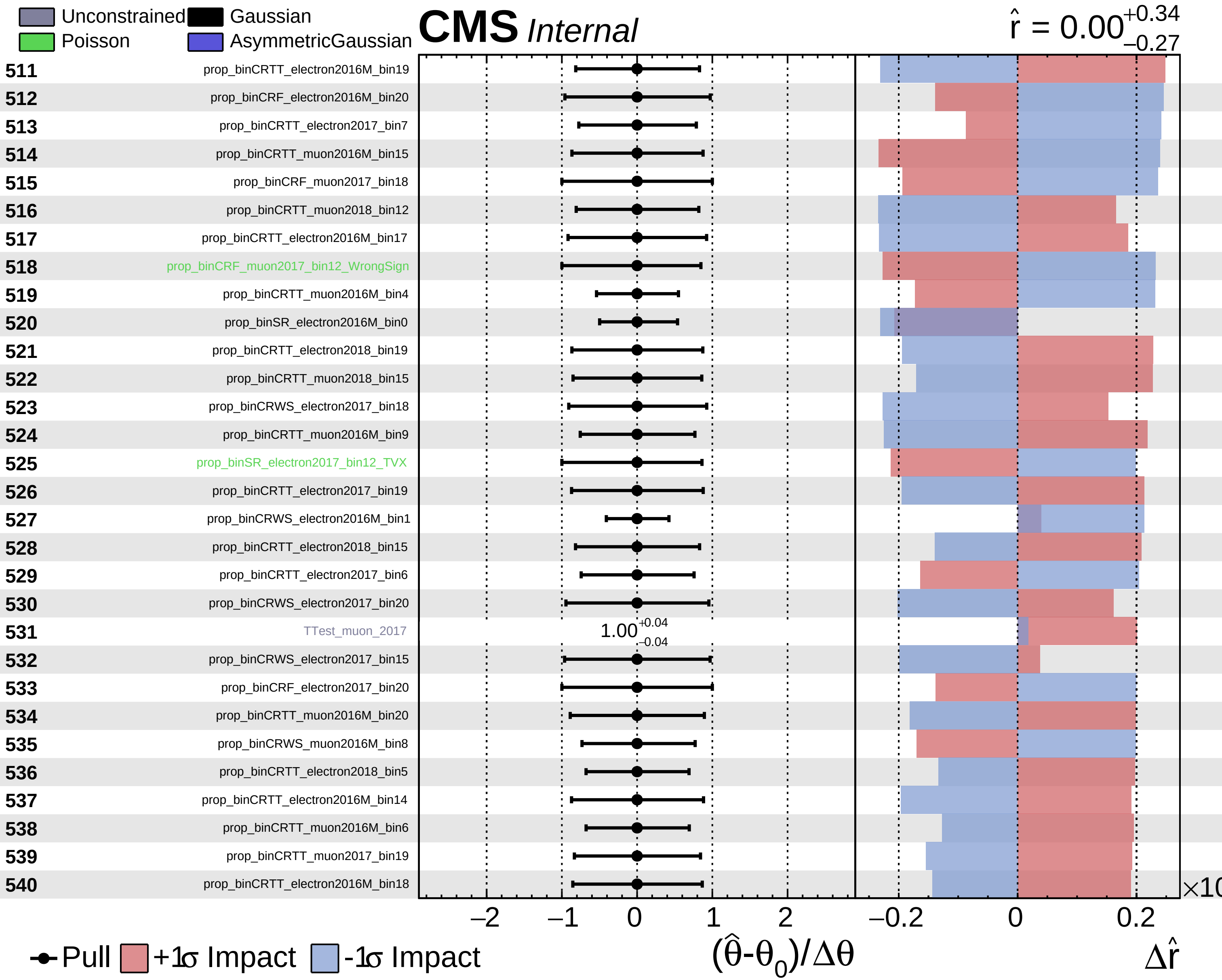


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
421 TTest_muon_2016M
1.00 +0.05 −0.05
422 prop_binCRTT_electron2016M_bin10
423 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
424 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
425 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
426 prop_binCRTT_electron2018_bin18
427 prop_binCRTT_electron2016M_bin3
428 prop_binCRTT_electron2017_bin18
429 prop_binSR_electron2017_bin14_WrongSign
430 prop_binCRWS_muon2018_bin11
431 prop_binSR_electron2016M_bin16_ZZtoLep
432 prop_binCRWS_muon2018_bin16
433 prop_binCRTT_electron2017_bin10
434 prop_binCRF_muon2016M_bin13
435 prop_binCRTT_muon2016M_bin11
436 prop_binCRF_muon2018_bin1
437 prop_binSR_muon2017_bin13_WpWpJJ_QCD
438 prop_binSR_electron2017_bin14_WpWpJJ_QCD
439 prop_binCRWS_electron2018_bin6_Fake
440 prop_binSR_electron2017_bin12_WpWpJJ_QCD
441 TTest_electron_2016M
1.00 +0.06 −0.06
442 prop_binCRWS_electron2018_bin8_Fake
443 prop_binCRTT_electron2018_bin6
444 prop_binCRWS_electron2017_bin5
445 prop_binCRWS_electron2016M_bin9_Fake
446 prop_binCRTT_electron2018_bin11
447 prop_binCRTT_muon2018_bin8
448 prop_binSR_electron2017_bin19_WrongSign
449 prop_binCRTT_muon2017_bin12
450 prop_binCRTT_muon2017_bin14
×10
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

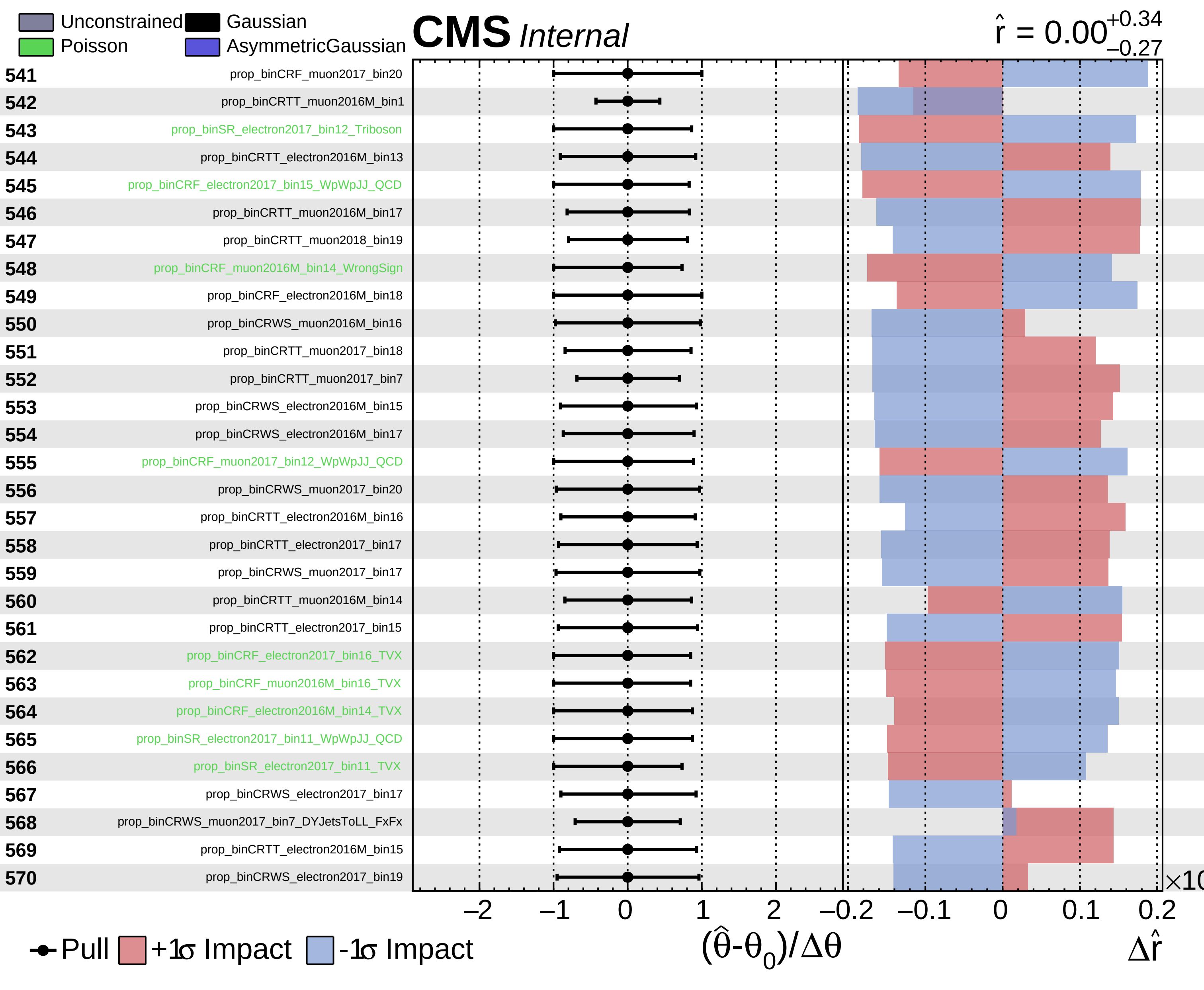


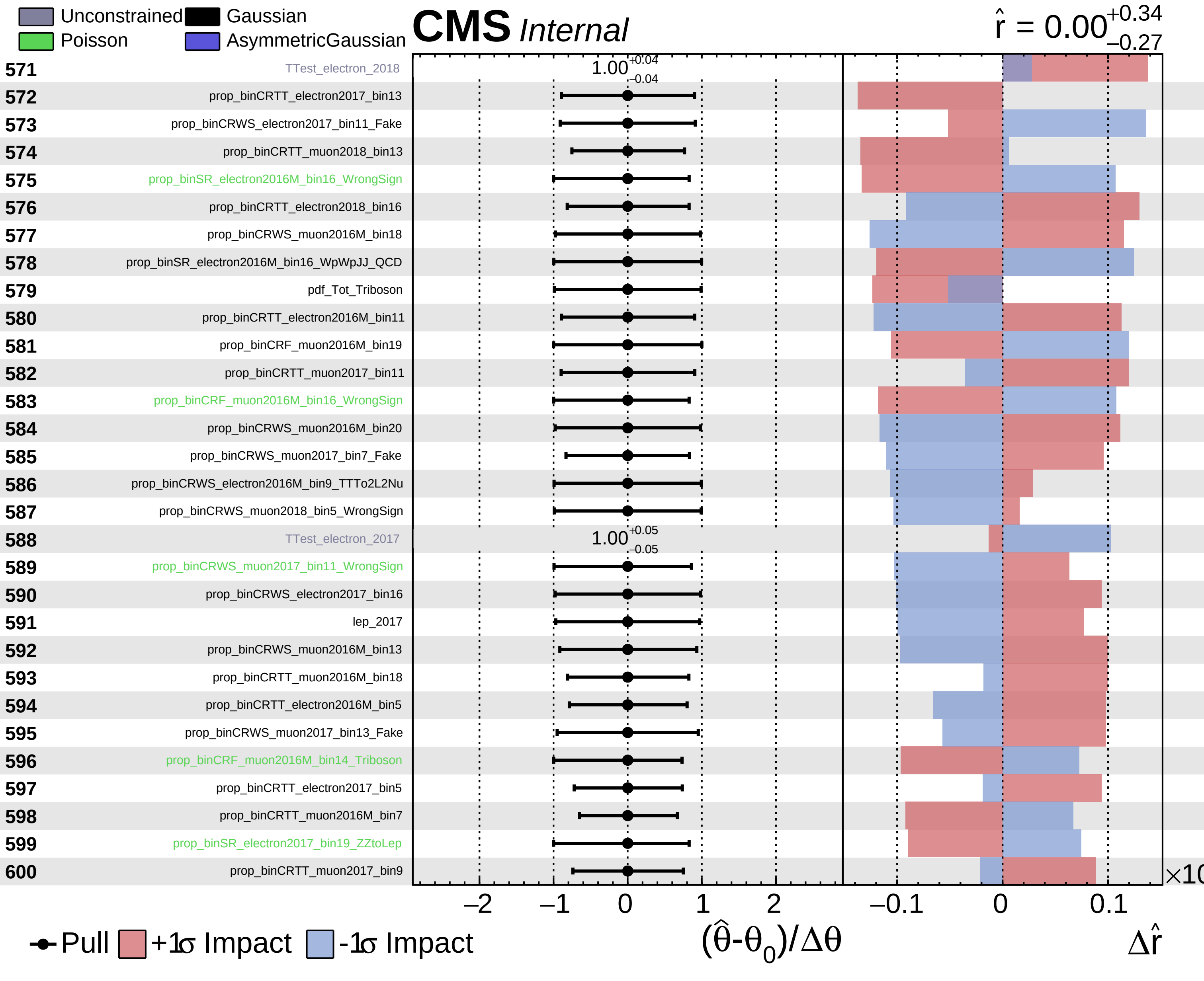
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
451 prop_binCRF_muon2018_bin19
452 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
453 prop_binCRTT_electron2016M_bin4
454 prop_binSR_electron2017_bin12_WrongSign
455 prop_binSR_electron2017_bin11_TTTo2L2Nu
456 prop_binCRWS_muon2016M_bin15
457 prop_binCRTT_muon2016M_bin12
458 prop_binCRTT_muon2016M_bin2
459 prop_binCRF_muon2018_bin16_TVX
460 prop_binCRWS_muon2018_bin19
461 prop_binCRF_electron2017_bin1
462 prop_binCRF_electron2017_bin18
463 prop_binCRTT_electron2017_bin9
464 prop_binCRWS_electron2018_bin4
465 ISR
466 prop_binCRTT_muon2018_bin11
467 prop_binCRTT_muon2016M_bin0
468 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
469 prop_binCRWS_electron2018_bin16_DYJetsToLL_FxFx
470 prop_binCRWS_muon2017_bin11_Fake
471 prop_binCRTT_electron2016M_bin2
472 prop_binCRTT_muon2016M_bin3
473 prop_binCRTT_electron2018_bin7
474 prop_binCRF_electron2016M_bin16
475 prop_binCRF_electron2017_bin19
476 prop_binCRF_electron2016M_bin17
477 tau_vsjet_2016M
478 prop_binCRTT_muon2016M_bin8
479 prop_binCRF_muon2018_bin16_WrongSign
480 prop_binSR_electron2017_bin19_Triboson
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
481 prop_binCRTT_muon2016M_bin13
482 TTest_muon_2018
1.00 +0.04 −0.04
483 prop_binCRWS_muon2017_bin16
484 prop_binCRTT_electron2018_bin17
485 prop_binCRTT_muon2018_bin16
486 prop_binCRF_muon2018_bin20
487 prop_binCRTT_muon2016M_bin5
488 prop_binCRTT_muon2017_bin16
489 prop_binCRF_muon2017_bin12_TVX
490 prop_binCRWS_electron2016M_bin4
491 prop_binCRTT_muon2017_bin13
492 prop_binCRTT_electron2017_bin12
493 prop_binCRTT_electron2017_bin8
494 prop_binCRF_muon2016M_bin14_WpWpJJ_QCD
495 prop_binCRWS_muon2017_bin5
496 prop_binCRWS_electron2017_bin14_Fake
497 prop_binCRTT_electron2017_bin14
498 prop_binCRF_muon2018_bin16_WpWpJJ_QCD
499 prop_binCRF_electron2017_bin16_WrongSign
500 prop_binCRTT_electron2017_bin16
501 prop_binSR_electron2017_bin0
502 prop_binCRTT_muon2017_bin3
503 prop_binCRWS_muon2016M_bin19
504 prop_binCRWS_electron2018_bin18
505 prop_binCRTT_electron2016M_bin20
506 prop_binCRF_electron2018_bin2
507 prop_binSR_electron2017_bin14_Triboson
508 prop_binSR_electron2016M_bin16_TVX
509 prop_binCRWS_electron2016M_bin7
510 prop_binCRTT_electron2016M_bin8
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

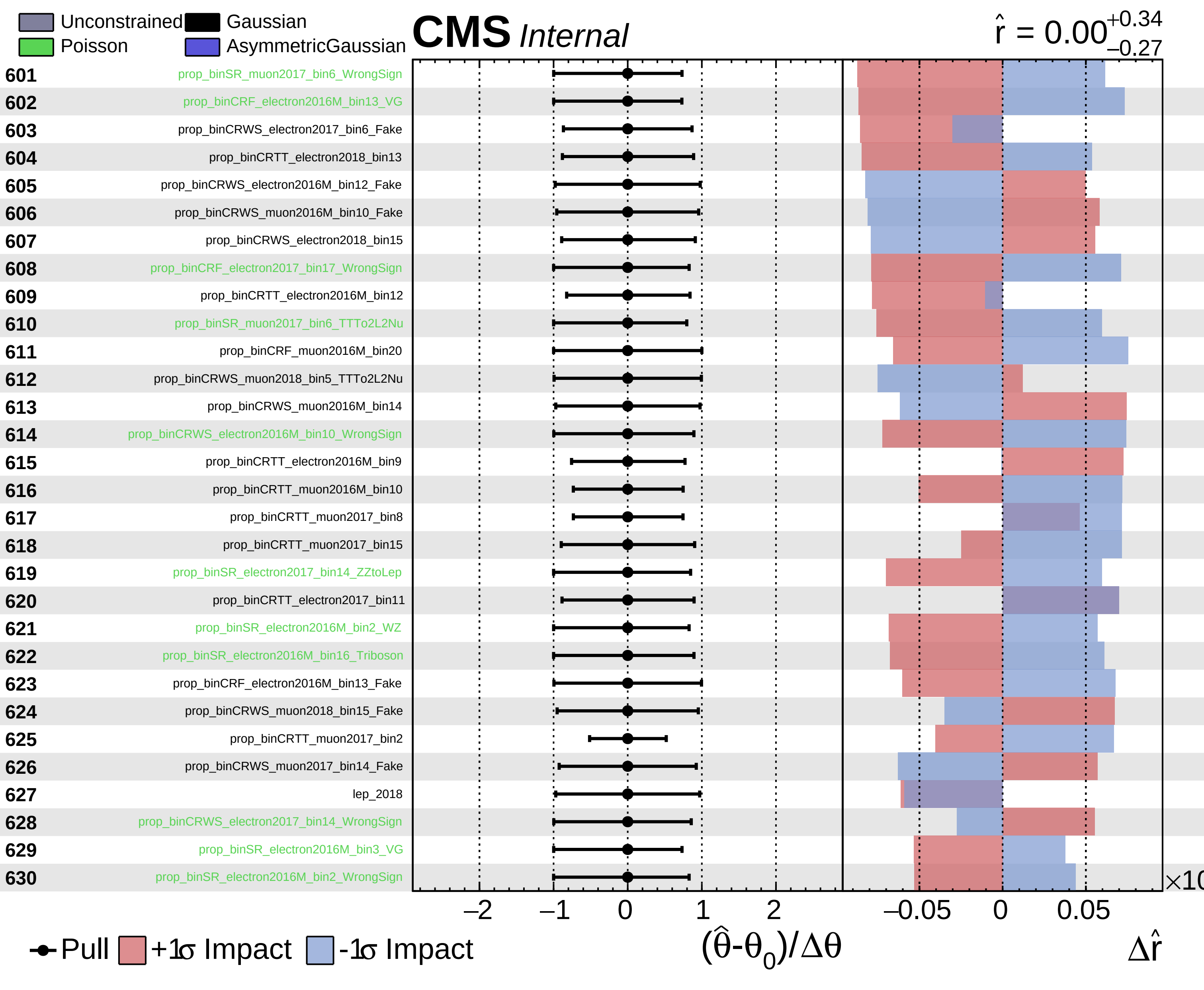


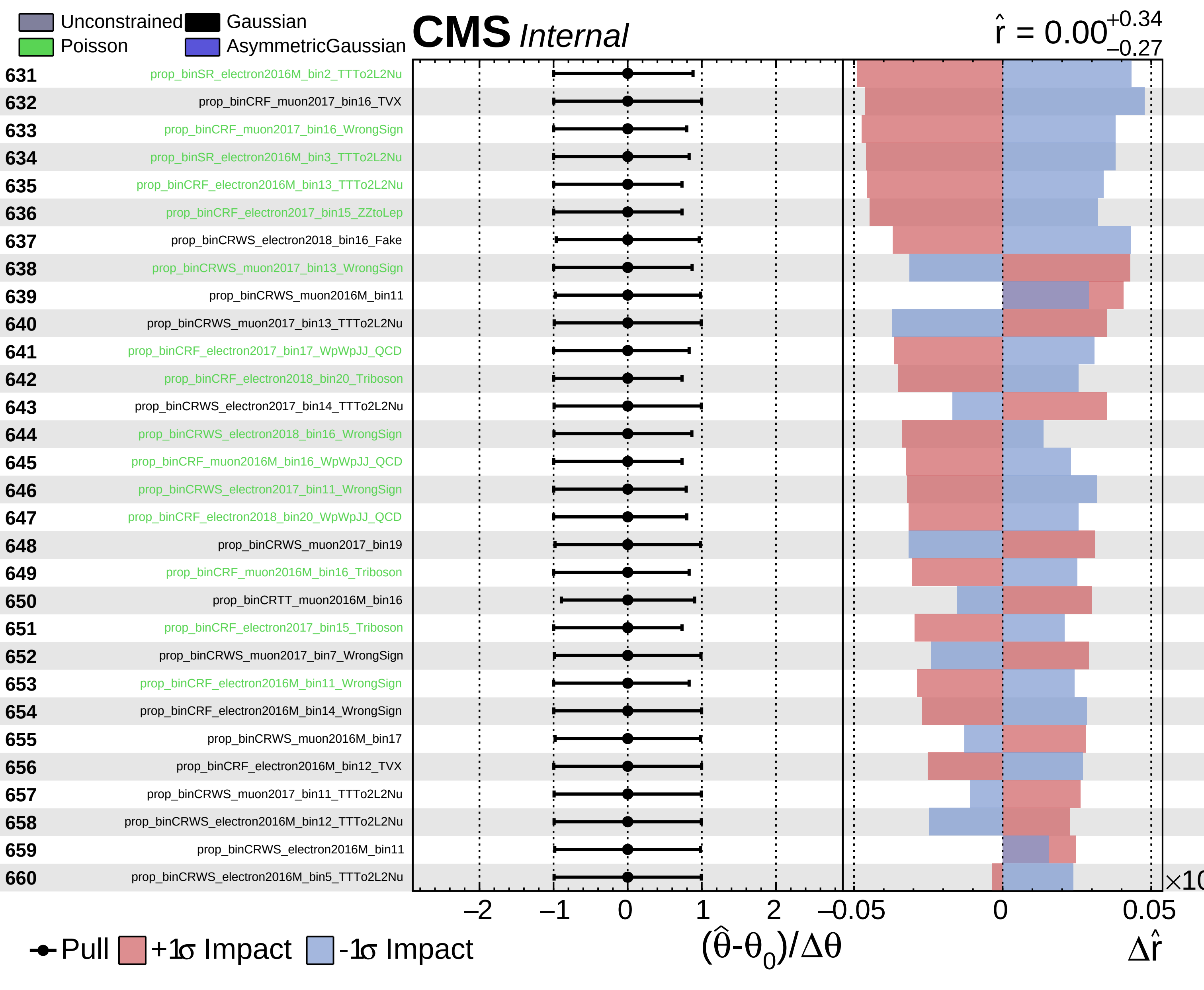
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
511 prop_binCRTT_electron2016M_bin19
512 prop_binCRF_electron2016M_bin20
513 prop_binCRTT_electron2017_bin7
514 prop_binCRTT_muon2016M_bin15
515 prop_binCRF_muon2017_bin18
516 prop_binCRTT_muon2018_bin12
517 prop_binCRTT_electron2016M_bin17
518 prop_binCRF_muon2017_bin12_WrongSign
519 prop_binCRTT_muon2016M_bin4
520 prop_binSR_electron2016M_bin0
521 prop_binCRTT_electron2018_bin19
522 prop_binCRTT_muon2018_bin15
523 prop_binCRWS_electron2017_bin18
524 prop_binCRTT_muon2016M_bin9
525 prop_binSR_electron2017_bin12_TVX
526 prop_binCRTT_electron2017_bin19
527 prop_binCRWS_electron2016M_bin1
528 prop_binCRTT_electron2018_bin15
529 prop_binCRTT_electron2017_bin6
530 prop_binCRWS_electron2017_bin20
531 TTest_muon_2017
1.00 +0.04 −0.04
532 prop_binCRWS_electron2017_bin15
533 prop_binCRF_electron2017_bin20
534 prop_binCRTT_muon2016M_bin20
535 prop_binCRWS_muon2016M_bin8
536 prop_binCRTT_electron2018_bin5
537 prop_binCRTT_electron2016M_bin14
538 prop_binCRTT_muon2016M_bin6
539 prop_binCRTT_muon2017_bin19
540 prop_binCRTT_electron2016M_bin18
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

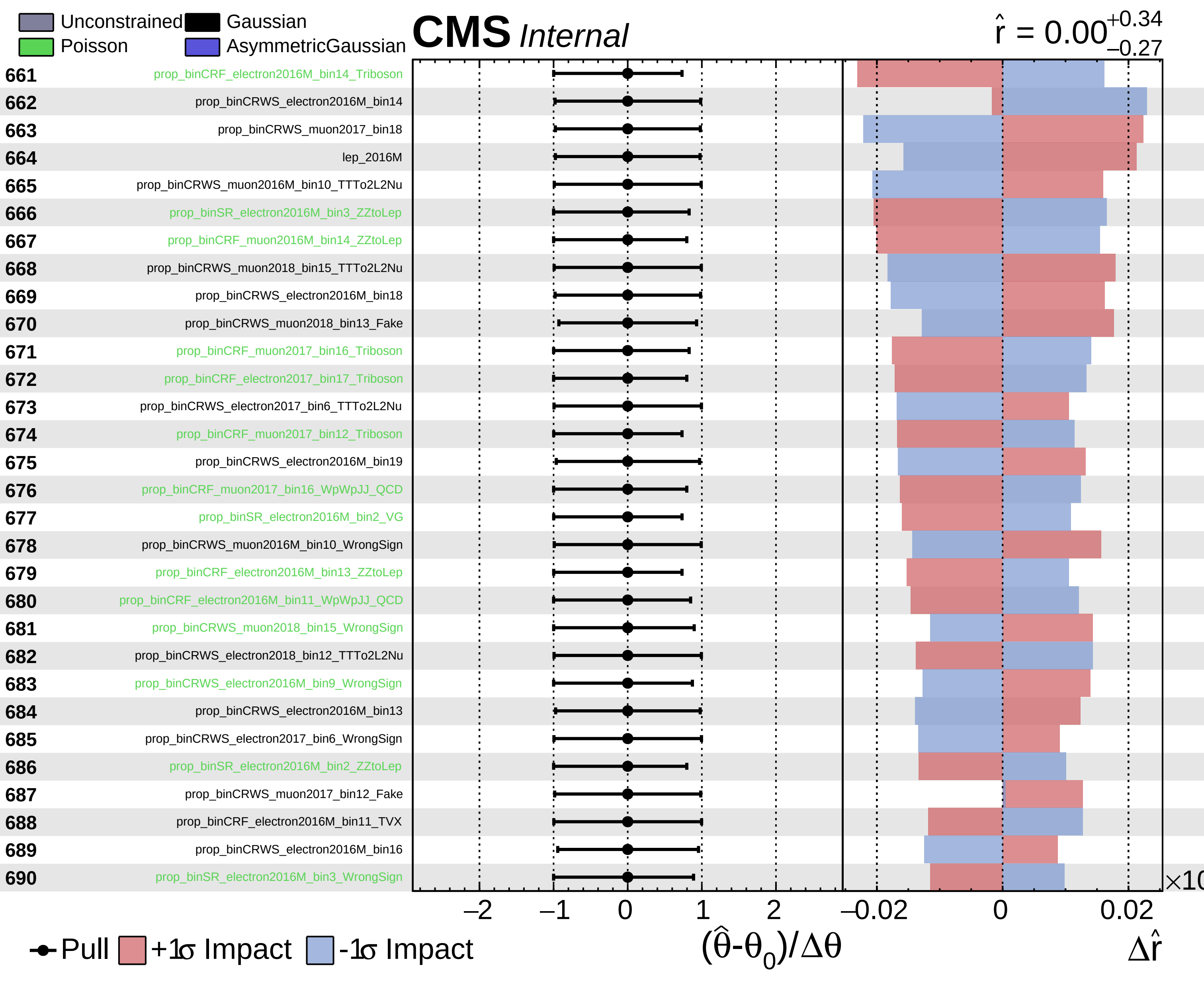

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
541 prop_binCRF_muon2017_bin20
542 prop_binCRTT_muon2016M_bin1
543 prop_binSR_electron2017_bin12_Triboson
544 prop_binCRTT_electron2016M_bin13
545 prop_binCRF_electron2017_bin15_WpWpJJ_QCD
546 prop_binCRTT_muon2016M_bin17
547 prop_binCRTT_muon2018_bin19
548 prop_binCRF_muon2016M_bin14_WrongSign
549 prop_binCRF_electron2016M_bin18
550 prop_binCRWS_muon2016M_bin16
551 prop_binCRTT_muon2017_bin18
552 prop_binCRTT_muon2017_bin7
553 prop_binCRWS_electron2016M_bin15
554 prop_binCRWS_electron2016M_bin17
555 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
556 prop_binCRWS_muon2017_bin20
557 prop_binCRTT_electron2016M_bin16
558 prop_binCRTT_electron2017_bin17
559 prop_binCRWS_muon2017_bin17
560 prop_binCRTT_muon2016M_bin14
561 prop_binCRTT_electron2017_bin15
562 prop_binCRF_electron2017_bin16_TVX
563 prop_binCRF_muon2016M_bin16_TVX
564 prop_binCRF_electron2016M_bin14_TVX
565 prop_binSR_electron2017_bin11_WpWpJJ_QCD
566 prop_binSR_electron2017_bin11_TVX
567 prop_binCRWS_electron2017_bin17
568 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
569 prop_binCRTT_electron2016M_bin15
570 prop_binCRWS_electron2017_bin19
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

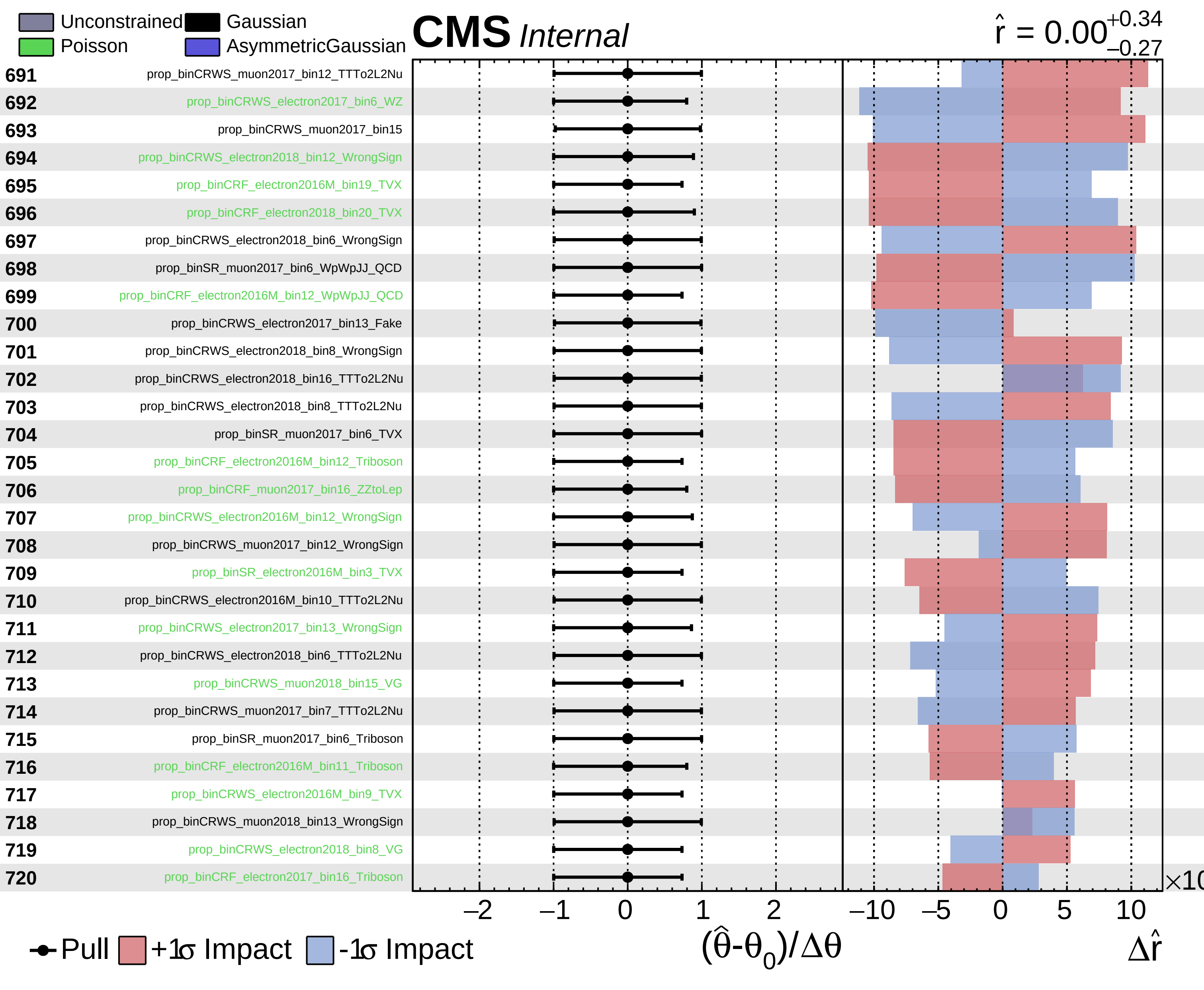


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
571 TTest_electron_2018
$1.00^{+0.04}_{-0.04}$
572 prop_binCRTT_electron2017_bin13
573 prop_binCRWS_electron2017_bin11_Fake
574 prop_binCRTT_muon2018_bin13
575 prop_binSR_electron2016M_bin16_WrongSign
576 prop_binCRTT_electron2018_bin16
577 prop_binCRWS_muon2016M_bin18
578 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
579 pdf_Tot_Triboson
580 prop_binCRTT_electron2016M_bin11
581 prop_binCRF_muon2016M_bin19
582 prop_binCRTT_muon2017_bin11
583 prop_binCRF_muon2016M_bin16_WrongSign
584 prop_binCRWS_muon2016M_bin20
585 prop_binCRWS_muon2017_bin7_Fake
586 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
587 prop_binCRWS_muon2018_bin5_WrongSign
588 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
589 prop_binCRWS_muon2017_bin11_WrongSign
590 prop_binCRWS_electron2017_bin16
591 lep_2017
592 prop_binCRWS_muon2016M_bin13
593 prop_binCRTT_muon2016M_bin18
594 prop_binCRTT_electron2016M_bin5
595 prop_binCRWS_muon2017_bin13_Fake
596 prop_binCRF_muon2016M_bin14_Triboson
597 prop_binCRTT_electron2017_bin5
598 prop_binCRTT_muon2016M_bin7
599 prop_binSR_electron2017_bin19_ZZtoLep
600 prop_binCRTT_muon2017_bin9
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

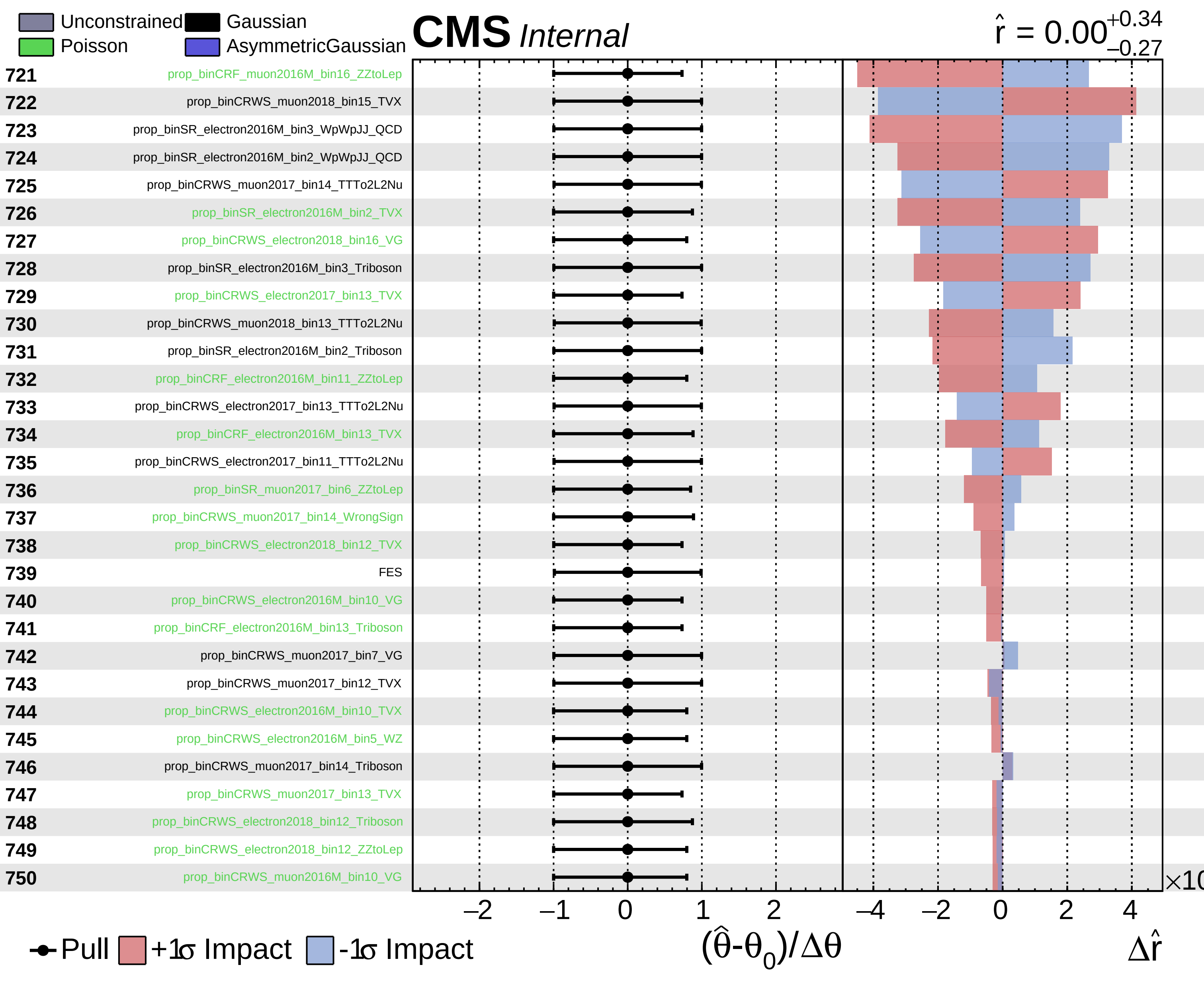

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
601 prop_binSR_muon2017_bin6_WrongSign
602 prop_binCRF_electron2016M_bin13_VG
603 prop_binCRWS_electron2017_bin6_Fake
604 prop_binCRTT_electron2018_bin13
605 prop_binCRWS_electron2016M_bin12_Fake
606 prop_binCRWS_muon2016M_bin10_Fake
607 prop_binCRWS_electron2018_bin15
608 prop_binCRF_electron2017_bin17_WrongSign
609 prop_binCRTT_electron2016M_bin12
610 prop_binSR_muon2017_bin6_TTTo2L2Nu
611 prop_binCRF_muon2016M_bin20
612 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
613 prop_binCRWS_muon2016M_bin14
614 prop_binCRWS_electron2016M_bin10_WrongSign
615 prop_binCRTT_electron2016M_bin9
616 prop_binCRTT_muon2016M_bin10
617 prop_binCRTT_muon2017_bin8
618 prop_binCRTT_muon2017_bin15
619 prop_binSR_electron2017_bin14_ZZtoLep
620 prop_binCRTT_electron2017_bin11
621 prop_binSR_electron2016M_bin2_WZ
622 prop_binSR_electron2016M_bin16_Triboson
623 prop_binCRF_electron2016M_bin13_Fake
624 prop_binCRWS_muon2018_bin15_Fake
625 prop_binCRTT_muon2017_bin2
626 prop_binCRWS_muon2017_bin14_Fake
627 lep_2018
628 prop_binCRWS_electron2017_bin14_WrongSign
629 prop_binSR_electron2016M_bin3_VG
630 prop_binSR_electron2016M_bin2_WrongSign
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
631 prop_binSR_electron2016M_bin2_TTTo2L2Nu
632 prop_binCRF_muon2017_bin16_TVX
633 prop_binCRF_muon2017_bin16_WrongSign
634 prop_binSR_electron2016M_bin3_TTTo2L2Nu
635 prop_binCRF_electron2016M_bin13_TTTo2L2Nu
636 prop_binCRF_electron2017_bin15_ZZtoLep
637 prop_binCRWS_electron2018_bin16_Fake
638 prop_binCRWS_muon2017_bin13_WrongSign
639 prop_binCRWS_muon2016M_bin11
640 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
641 prop_binCRF_electron2017_bin17_WpWpJJ_QCD
642 prop_binCRF_electron2018_bin20_Triboson
643 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
644 prop_binCRWS_electron2018_bin16_WrongSign
645 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
646 prop_binCRWS_electron2017_bin11_WrongSign
647 prop_binCRF_electron2018_bin20_WpWpJJ_QCD
648 prop_binCRWS_muon2017_bin19
649 prop_binCRF_muon2016M_bin16_Triboson
650 prop_binCRTT_muon2016M_bin16
651 prop_binCRF_electron2017_bin15_Triboson
652 prop_binCRWS_muon2017_bin7_WrongSign
653 prop_binCRF_electron2016M_bin11_WrongSign
654 prop_binCRF_electron2016M_bin14_WrongSign
655 prop_binCRWS_muon2016M_bin17
656 prop_binCRF_electron2016M_bin12_TVX
657 prop_binCRWS_muon2017_bin11_TTTo2L2Nu
658 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
659 prop_binCRWS_electron2016M_bin11
660 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

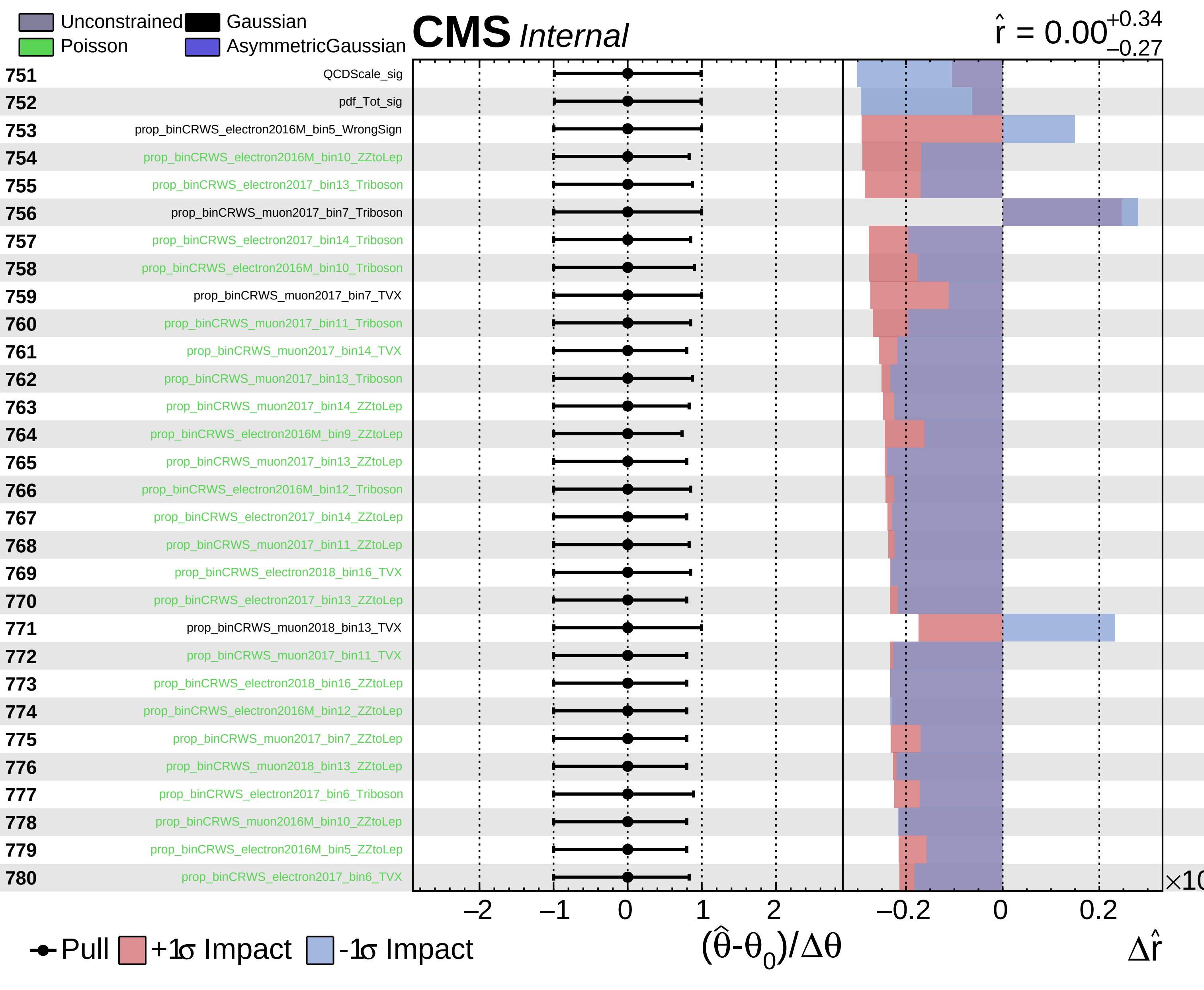

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
661 prop_binCRF_electron2016M_bin14_Triboson
662 prop_binCRWS_electron2016M_bin14
663 prop_binCRWS_muon2017_bin18
664 lep_2016M
665 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
666 prop_binSR_electron2016M_bin3_ZZtoLep
667 prop_binCRF_muon2016M_bin14_ZZtoLep
668 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
669 prop_binCRWS_electron2016M_bin18
670 prop_binCRWS_muon2018_bin13_Fake
671 prop_binCRF_muon2017_bin16_Triboson
672 prop_binCRF_electron2017_bin17_Triboson
673 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
674 prop_binCRF_muon2017_bin12_Triboson
675 prop_binCRWS_electron2016M_bin19
676 prop_binCRF_muon2017_bin16_WpWpJJ_QCD
677 prop_binSR_electron2016M_bin2_VG
678 prop_binCRWS_muon2016M_bin10_WrongSign
679 prop_binCRF_electron2016M_bin13_ZZtoLep
680 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
681 prop_binCRWS_muon2018_bin15_WrongSign
682 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
683 prop_binCRWS_electron2016M_bin9_WrongSign
684 prop_binCRWS_electron2016M_bin13
685 prop_binCRWS_electron2017_bin6_WrongSign
686 prop_binSR_electron2016M_bin2_ZZtoLep
687 prop_binCRWS_muon2017_bin12_Fake
688 prop_binCRF_electron2016M_bin11_TVX
689 prop_binCRWS_electron2016M_bin16
690 prop_binSR_electron2016M_bin3_WrongSign
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

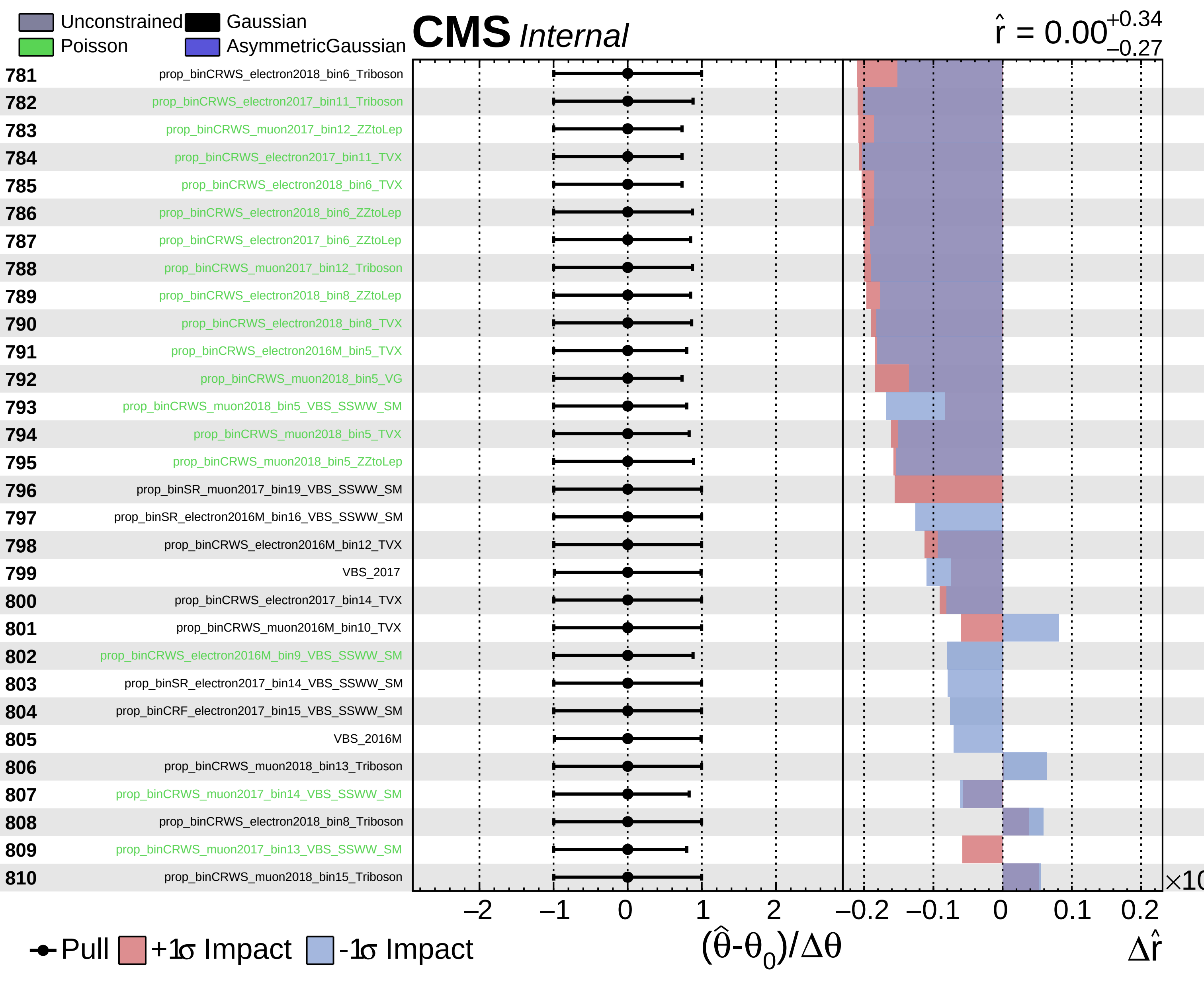


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
691 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
692 prop_binCRWS_electron2017_bin6_WZ
693 prop_binCRWS_muon2017_bin15
694 prop_binCRWS_electron2018_bin12_WrongSign
695 prop_binCRF_electron2016M_bin19_TVX
696 prop_binCRF_electron2018_bin20_TVX
697 prop_binCRWS_electron2018_bin6_WrongSign
698 prop_binSR_muon2017_bin6_WpWpJJ_QCD
699 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
700 prop_binCRWS_electron2017_bin13_Fake
701 prop_binCRWS_electron2018_bin8_WrongSign
702 prop_binCRWS_electron2018_bin16_TTTo2L2Nu
703 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
704 prop_binSR_muon2017_bin6_TVX
705 prop_binCRF_electron2016M_bin12_Triboson
706 prop_binCRF_muon2017_bin16_ZZtoLep
707 prop_binCRWS_electron2016M_bin12_WrongSign
708 prop_binCRWS_muon2017_bin12_WrongSign
709 prop_binSR_electron2016M_bin3_TVX
710 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
711 prop_binCRWS_electron2017_bin13_WrongSign
712 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
713 prop_binCRWS_muon2018_bin15_VG
714 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
715 prop_binSR_muon2017_bin6_Triboson
716 prop_binCRF_electron2016M_bin11_Triboson
717 prop_binCRWS_electron2016M_bin9_TVX
718 prop_binCRWS_muon2018_bin13_WrongSign
719 prop_binCRWS_electron2018_bin8_VG
720 prop_binCRF_electron2017_bin16_Triboson
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

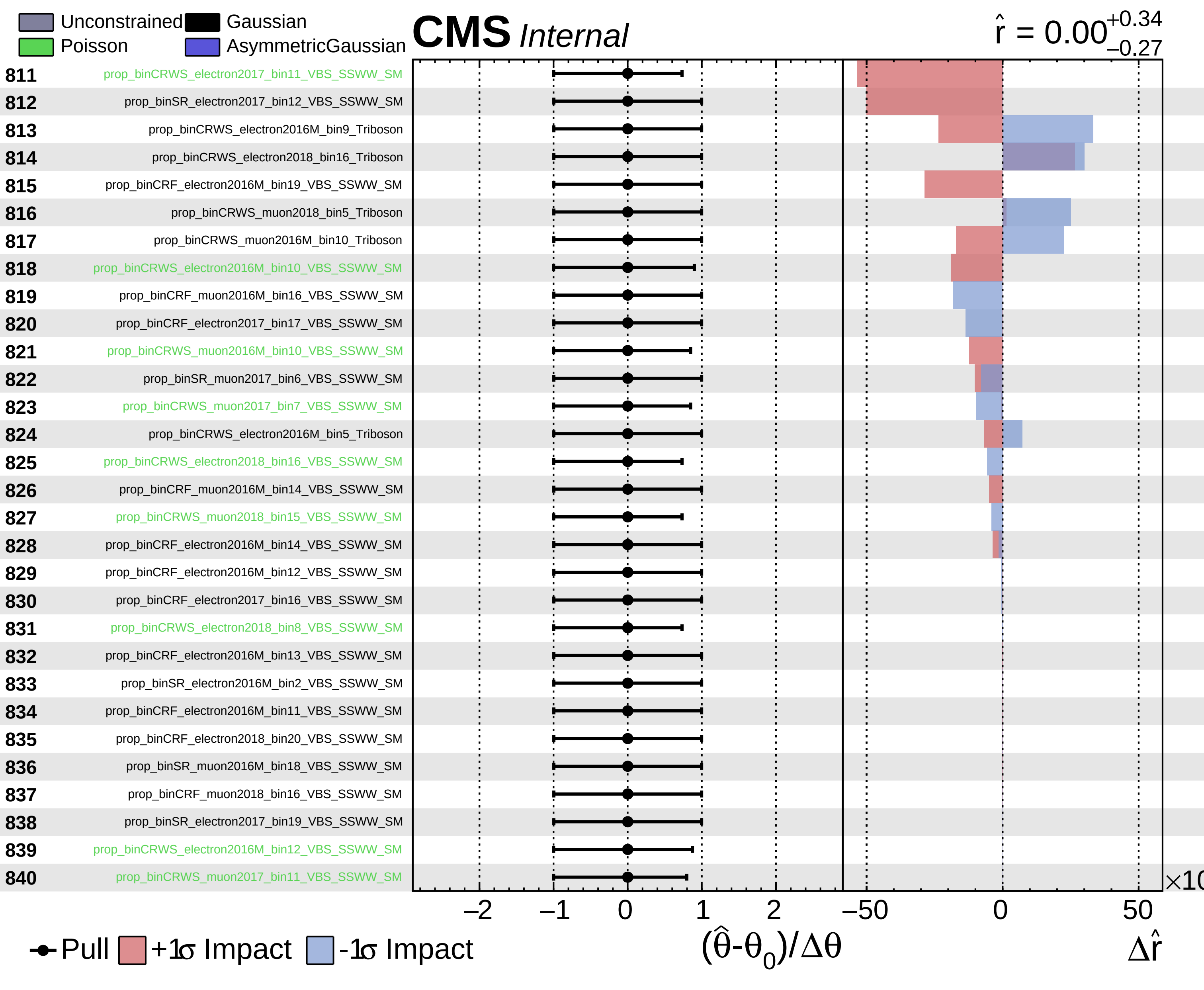


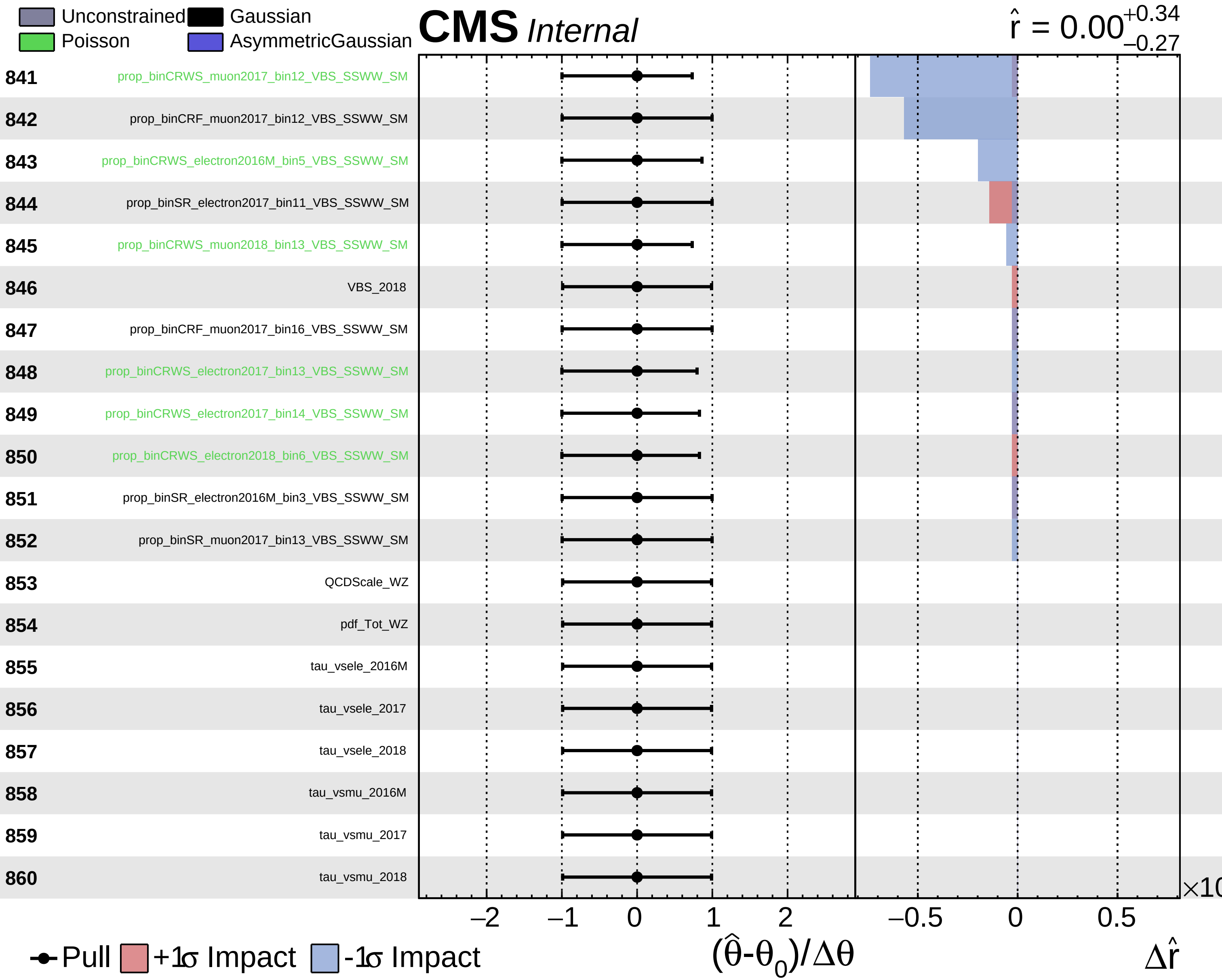
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
721 prop_binCRF_muon2016M_bin16_ZZtoLep
722 prop_binCRWS_muon2018_bin15_TVX
723 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
724 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
725 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
726 prop_binSR_electron2016M_bin2_TVX
727 prop_binCRWS_electron2018_bin16_VG
728 prop_binSR_electron2016M_bin3_Triboson
729 prop_binCRWS_electron2017_bin13_TVX
730 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
731 prop_binSR_electron2016M_bin2_Triboson
732 prop_binCRF_electron2016M_bin11_ZZtoLep
733 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
734 prop_binCRF_electron2016M_bin13_TVX
735 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
736 prop_binSR_muon2017_bin6_ZZtoLep
737 prop_binCRWS_muon2017_bin14_WrongSign
738 prop_binCRWS_electron2018_bin12_TVX
739 FES
740 prop_binCRWS_electron2016M_bin10_VG
741 prop_binCRF_electron2016M_bin13_Triboson
742 prop_binCRWS_muon2017_bin7_VG
743 prop_binCRWS_muon2017_bin12_TVX
744 prop_binCRWS_electron2016M_bin10_TVX
745 prop_binCRWS_electron2016M_bin5_WZ
746 prop_binCRWS_muon2017_bin14_Triboson
747 prop_binCRWS_muon2017_bin13_TVX
748 prop_binCRWS_electron2018_bin12_Triboson
749 prop_binCRWS_electron2018_bin12_ZZtoLep
750 prop_binCRWS_muon2016M_bin10_VG
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
751 QCDScale_sig
752 pdf_Tot_sig
753 prop_binCRWS_electron2016M_bin5_WrongSign
754 prop_binCRWS_electron2016M_bin10_ZZtoLep
755 prop_binCRWS_electron2017_bin13_Triboson
756 prop_binCRWS_muon2017_bin7_Triboson
757 prop_binCRWS_electron2017_bin14_Triboson
758 prop_binCRWS_electron2016M_bin10_Triboson
759 prop_binCRWS_muon2017_bin7_TVX
760 prop_binCRWS_muon2017_bin11_Triboson
761 prop_binCRWS_muon2017_bin14_TVX
762 prop_binCRWS_muon2017_bin13_Triboson
763 prop_binCRWS_muon2017_bin14_ZZtoLep
764 prop_binCRWS_electron2016M_bin9_ZZtoLep
765 prop_binCRWS_muon2017_bin13_ZZtoLep
766 prop_binCRWS_electron2016M_bin12_Triboson
767 prop_binCRWS_electron2017_bin14_ZZtoLep
768 prop_binCRWS_muon2017_bin11_ZZtoLep
769 prop_binCRWS_electron2018_bin16_TVX
770 prop_binCRWS_electron2017_bin13_ZZtoLep
771 prop_binCRWS_muon2018_bin13_TVX
772 prop_binCRWS_muon2017_bin11_TVX
773 prop_binCRWS_electron2018_bin16_ZZtoLep
774 prop_binCRWS_electron2016M_bin12_ZZtoLep
775 prop_binCRWS_muon2017_bin7_ZZtoLep
776 prop_binCRWS_muon2018_bin13_ZZtoLep
777 prop_binCRWS_electron2017_bin6_Triboson
778 prop_binCRWS_muon2016M_bin10_ZZtoLep
779 prop_binCRWS_electron2016M_bin5_ZZtoLep
780 prop_binCRWS_electron2017_bin6_TVX
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
781 prop_binCRWS_electron2018_bin6_Triboson
782 prop_binCRWS_electron2017_bin11_Triboson
783 prop_binCRWS_muon2017_bin12_ZZtoLep
784 prop_binCRWS_electron2017_bin11_TVX
785 prop_binCRWS_electron2018_bin6_TVX
786 prop_binCRWS_electron2018_bin6_ZZtoLep
787 prop_binCRWS_electron2017_bin6_ZZtoLep
788 prop_binCRWS_muon2017_bin12_Triboson
789 prop_binCRWS_electron2018_bin8_ZZtoLep
790 prop_binCRWS_electron2018_bin8_TVX
791 prop_binCRWS_electron2016M_bin5_TVX
792 prop_binCRWS_muon2018_bin5_VG
793 prop_binCRWS_muon2018_bin5_VBS_SSWW_SM
794 prop_binCRWS_muon2018_bin5_TVX
795 prop_binCRWS_muon2018_bin5_ZZtoLep
796 prop_binSR_muon2017_bin19_VBS_SSWW_SM
797 prop_binSR_electron2016M_bin16_VBS_SSWW_SM
798 prop_binCRWS_electron2016M_bin12_TVX
799 VBS_2017
800 prop_binCRWS_electron2017_bin14_TVX
801 prop_binCRWS_muon2016M_bin10_TVX
802 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
803 prop_binSR_electron2017_bin14_VBS_SSWW_SM
804 prop_binCRF_electron2017_bin15_VBS_SSWW_SM
805 VBS_2016M
806 prop_binCRWS_muon2018_bin13_Triboson
807 prop_binCRWS_muon2017_bin14_VBS_SSWW_SM
808 prop_binCRWS_electron2018_bin8_Triboson
809 prop_binCRWS_muon2017_bin13_VBS_SSWW_SM
810 prop_binCRWS_muon2018_bin15_Triboson
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.27
811 prop_binCRWS_electron2017_bin11_VBS_SSWW_SM
812 prop_binSR_electron2017_bin12_VBS_SSWW_SM
813 prop_binCRWS_electron2016M_bin9_Triboson
814 prop_binCRWS_electron2018_bin16_Triboson
815 prop_binCRF_electron2016M_bin19_VBS_SSWW_SM
816 prop_binCRWS_muon2018_bin5_Triboson
817 prop_binCRWS_muon2016M_bin10_Triboson
818 prop_binCRWS_electron2016M_bin10_VBS_SSWW_SM
819 prop_binCRF_muon2016M_bin16_VBS_SSWW_SM
820 prop_binCRF_electron2017_bin17_VBS_SSWW_SM
821 prop_binCRWS_muon2016M_bin10_VBS_SSWW_SM
822 prop_binSR_muon2017_bin6_VBS_SSWW_SM
823 prop_binCRWS_muon2017_bin7_VBS_SSWW_SM
824 prop_binCRWS_electron2016M_bin5_Triboson
825 prop_binCRWS_electron2018_bin16_VBS_SSWW_SM
826 prop_binCRF_muon2016M_bin14_VBS_SSWW_SM
827 prop_binCRWS_muon2018_bin15_VBS_SSWW_SM
828 prop_binCRF_electron2016M_bin14_VBS_SSWW_SM
829 prop_binCRF_electron2016M_bin12_VBS_SSWW_SM
830 prop_binCRF_electron2017_bin16_VBS_SSWW_SM
831 prop_binCRWS_electron2018_bin8_VBS_SSWW_SM
832 prop_binCRF_electron2016M_bin13_VBS_SSWW_SM
833 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
834 prop_binCRF_electron2016M_bin11_VBS_SSWW_SM
835 prop_binCRF_electron2018_bin20_VBS_SSWW_SM
836 prop_binSR_muon2016M_bin18_VBS_SSWW_SM
837 prop_binCRF_muon2018_bin16_VBS_SSWW_SM
838 prop_binSR_electron2017_bin19_VBS_SSWW_SM
839 prop_binCRWS_electron2016M_bin12_VBS_SSWW_SM
840 prop_binCRWS_muon2017_bin11_VBS_SSWW_SM
−2 −1 0 1 2
−50 0 50
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.27}$
841 prop_binCRWS_muon2017_bin12_VBS_SSWW_SM
842 prop_binCRF_muon2017_bin12_VBS_SSWW_SM
843 prop_binCRWS_electron2016M_bin5_VBS_SSWW_SM
844 prop_binSR_electron2017_bin11_VBS_SSWW_SM
845 prop_binCRWS_muon2018_bin13_VBS_SSWW_SM
846 VBS_2018
847 prop_binCRF_muon2017_bin16_VBS_SSWW_SM
848 prop_binCRWS_electron2017_bin13_VBS_SSWW_SM
849 prop_binCRWS_electron2017_bin14_VBS_SSWW_SM
850 prop_binCRWS_electron2018_bin6_VBS_SSWW_SM
851 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
852 prop_binSR_muon2017_bin13_VBS_SSWW_SM
853 QCDScale_WZ
854 pdf_Tot_WZ
855 tau_vsele_2016M
856 tau_vsele_2017
857 tau_vsele_2018
858 tau_vsmu_2016M
859 tau_vsmu_2017
860 tau_vsmu_2018
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$